INCH-POUND

MIL-M-28787/350
28 April 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUFFER, THREE-STATE, DIGITAL, MODULE, KEY CODE YEZ

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital three-state buffer module. This specification covers one class of module. Class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.0652 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be YEZ.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned 1 through 20 in one row and 21 through 40 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

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TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation initial end-of-life	III	S	10 1	Inf Inf	10 1	Inf Inf	10 1	Inf Inf	MOhm MOhm
B1, B2	Continuity	IV	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B3, B4	Continuity	IV	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	V	S	--------Modules shall pass all remaining tests--------						
D1-D20	Ibd	VI	S	0	100	0	100	0	100	uA
E1-E16	Ios	VII	S	-225	-40	-225	-40	-225	-40	mA
F1-F20	Vic	VIII	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
G1-G16	Idzh	IX	X	-20	20	-20	20	-20	20	uA
H1-H16	Idzl	X	X	-20	20	-20	20	-20	20	uA
J1-J16	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K1-K16	Voh	XII	X	2.4	4.5	2.4	4.5	2.4	4.5	V
L1-L20	Iih	XIII	X	0	20	0	20	0	20	uA
M1-M20	Iil	XIV	X	-0.2	0.0	-0.2	0.0	-0.2	0.0	mA
N1	Icch	XV	X	5.4	54.0	5.4	54.0	5.4	54.0	mA
N2	Iccl	XV	X	9.2	92.0	9.2	92.0	9.2	92.0	mA
N3	Icchl	XV	X	10.8	108.0	10.8	108.0	10.8	108.0	mA
P1-P16	tPLH CL = 150 pF	XVI	S	5	29	5	29	5	29	nS
P17-P32	tPLH CL = 300 pF	XVI	Q	10	38	10	38	10	38	nS

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
Q1-Q16	tPHL CL = 150 pF	XVII	S	5	26	5	26	5	26	nS
Q17-Q32	tPHL CL = 300 pF	XVII	Q	10	32	10	32	10	32	nS
R1-R16	tPLZ CL = 50 pF	XVIII	Q	5	30	5	30	5	30	nS
S1-S16	tPZL CL = 150 pF	XIX	S	5	40	5	40	5	40	nS
T1-T16	tPHZ CL = 50 pF	XX	Q	5	35	5	35	5	35	nS
U1-U16	tPZH CL = 150 pF	XXI	S	5	40	5	40	5	40	nS
V1-V16	Xtalk1	XXII	Q	0	625	0	625	0	625	mV
W1-W16	Xtalkd	XXIII	Q	0	1000	0	1000	0	1000	mV
X1-X32	Xtalkd	XXIV	Q	0	33	0	33	0	33	nS
Y	PSTR	XXV	Q	0	400	0	400	0	400	mV
Z	PSFC	XXVI	S	3.03	6.38	3.03	6.38	3.03	6.38	uF
AA1-AA16	Cin	XXVII	Q	2	30	2	30	2	30	pF
AA17-AA20	Cin	XXVII	Q	2	35	2	35	2	35	pF
BB1-BB16	Cout	XXVIII	Q	2	35	2	35	2	35	pF

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/350-1 for class I. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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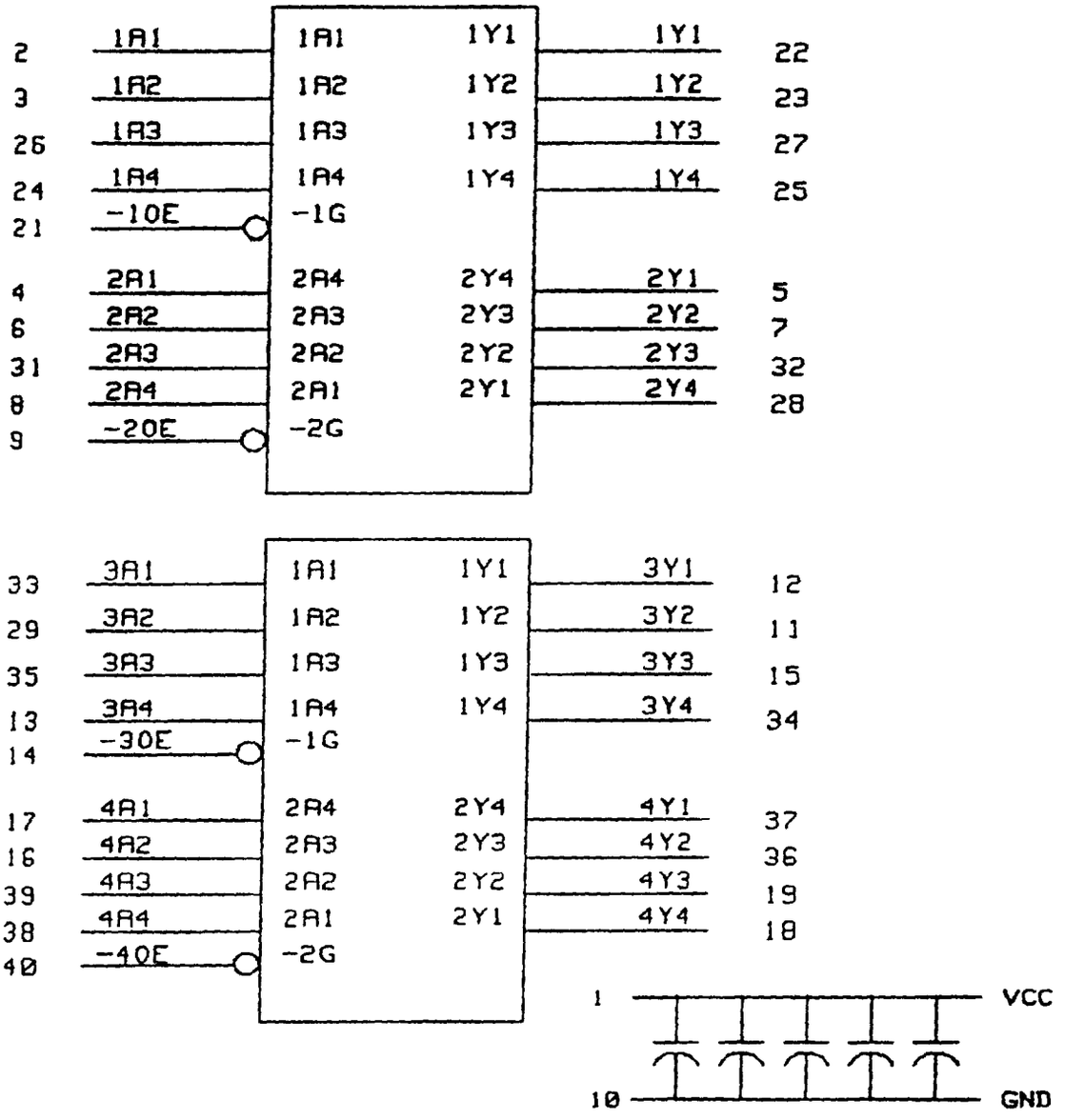


FIGURE 1. YEZ functional block diagram, three state buffer.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. Test conditions A and C through F are stress tests and will be performed only at 25 ± 5°C. If test conditions A and C through F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

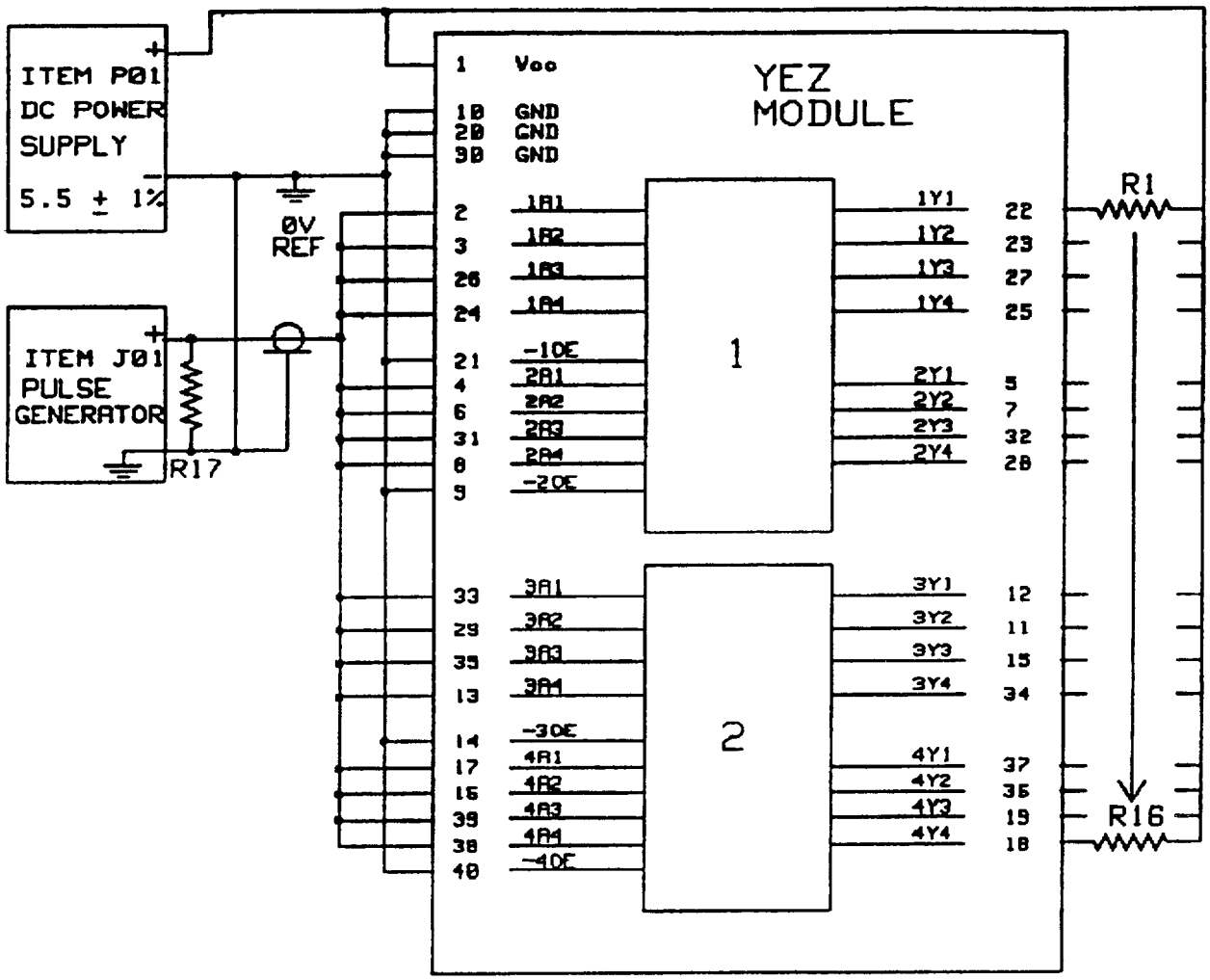
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and G through Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition Z, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTES:

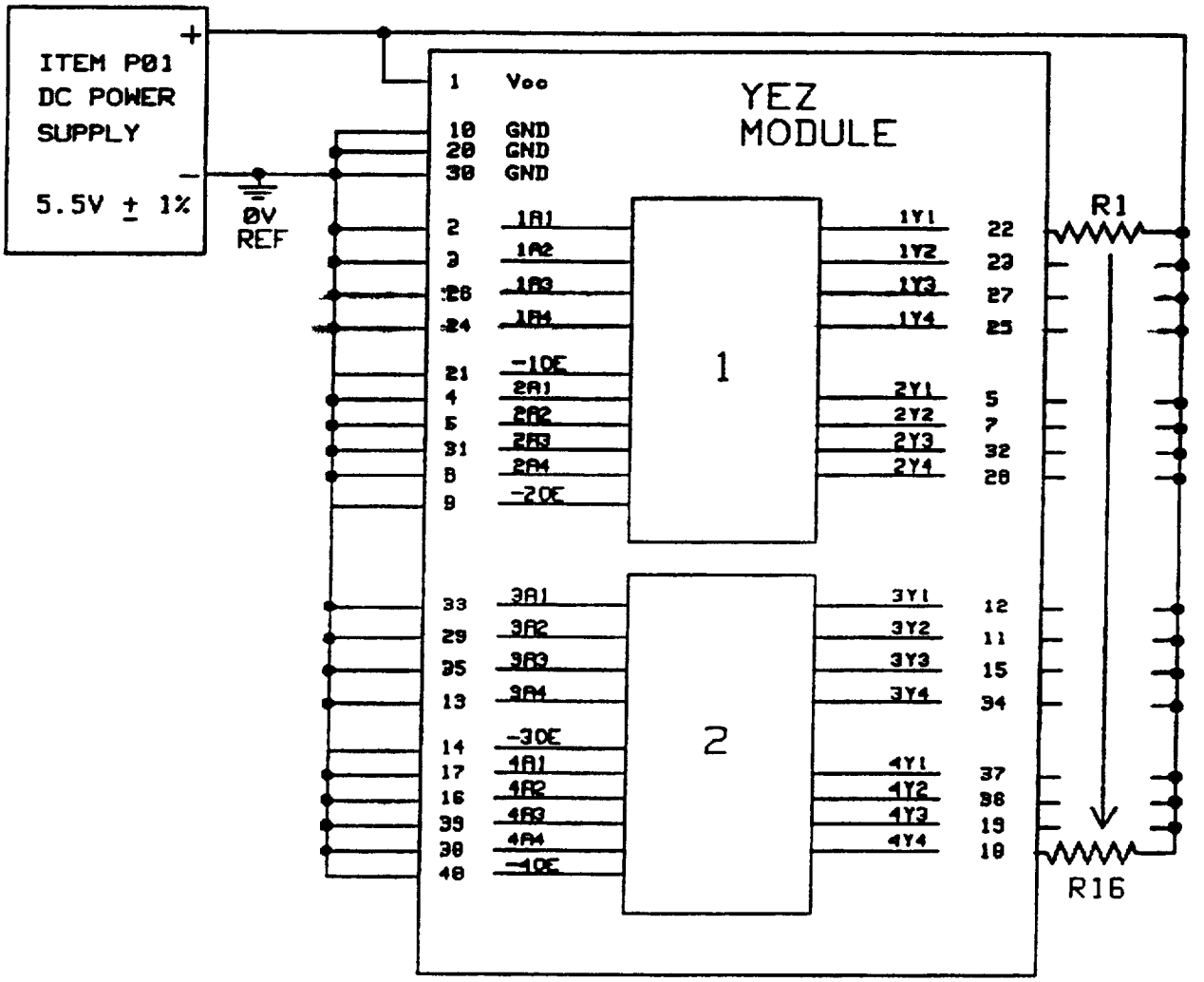
1. R1-R16 = 422 Ohm ± 1 percent, 1/4 W.

2. R17 = 51.1 Ohm ± 1 percent, 1/4 W.

3. Pulse Generator setup:
   PW = 500 ± 50 nS.
   PRR = 1.0 MHz ± 10 percent.
   PA = 4.0 ± 1.0 V.
   tTLH = tTHL = 50 ± 10 nS.

FIGURE 2. YEZ operating temperature and life test circuit.

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NOTE:

1. R1-R16 = 422 Ohm ± 1 percent, 1/4 W.

FIGURE 3. YEZ component temperature test circuit.

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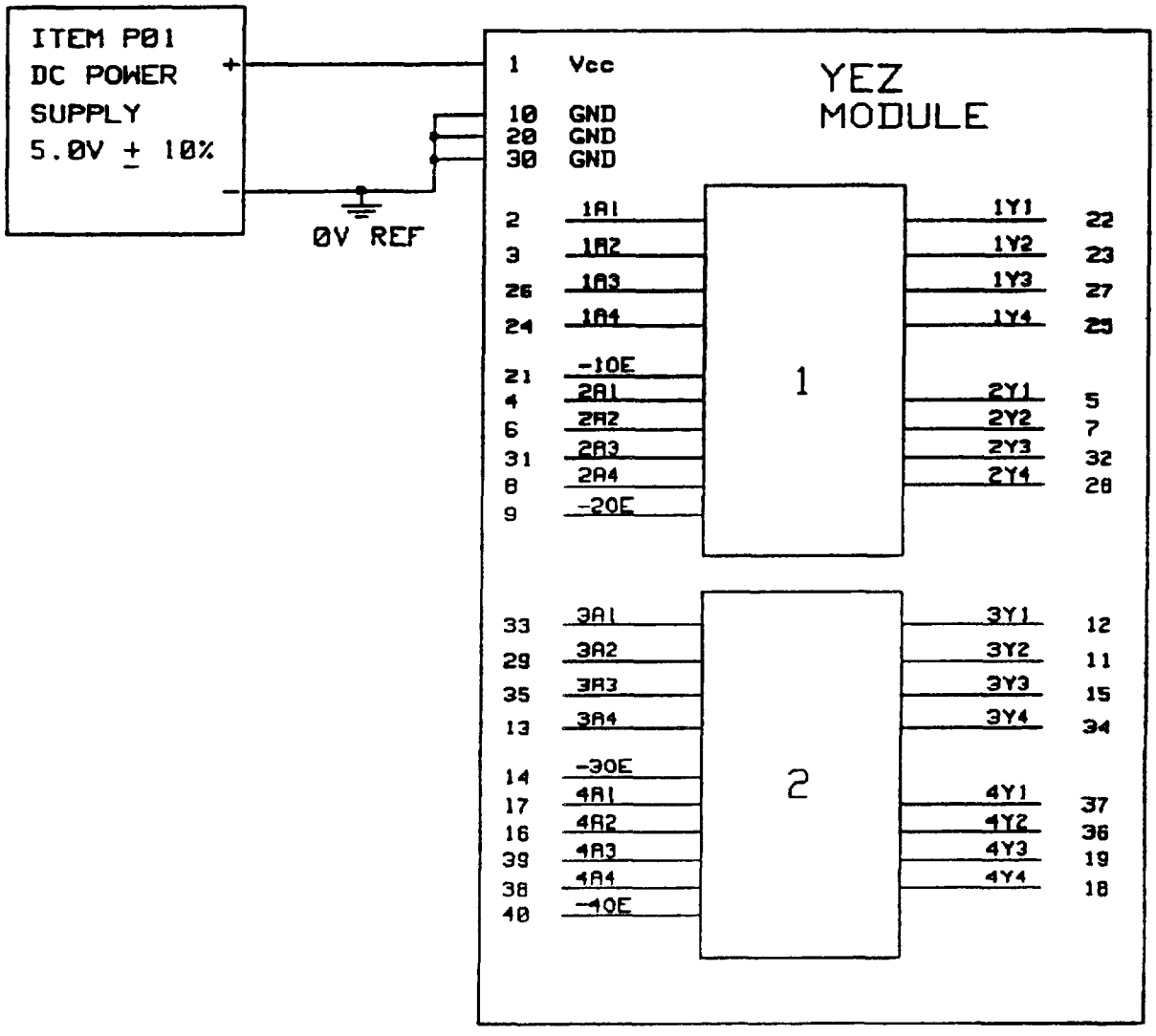


FIGURE 4. YEZ vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B and G through Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B and G through Z shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Icch = high level power supply current.
(g) Icchi = power supply current high impedance.
(h) Iccl = low level power supply current.
(i) Iil = input low level current.
(j) Iih = input high level current.
(k) Iin = input forcing current.
(l) Iol = output low level current.
(m) Ioh = output high level current.
(n) Ios = output short circuit current.
(o) Iozh = output high level leakage current.
(p) Iozl = output low level leakage current.
(q) NA = not applicable.
(r) PSFC = power supply filter capacitance.
(s) PSTA = power supply transient amplitude.
(t) tPHL = propagation high to low.
(u) tPHZ = propagation high to high impedance.
(v) tPLH = propagation low to high.
(w) tPZH = propagation high impedance to high level.
(x) Vcc = power supply voltage.
(y) Vic = input clamp diode voltage.
(z) Vil = input low level voltage.
(aa) Vih = input high level voltage.
(bb) Vin = input forcing voltage.
(cc) Voh = output high level voltage.
(dd) Vol = output low level voltage.
(ee) Vout = output forcing voltage.
(ff) Vpk = peak voltage.
(gg) Xtalkd = duration crosstalk.
(hh) Xtalkl = low level crosstalk.

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6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits in table I.

Supply voltage		5.0 V dc ± 10 percent
Maximum power dissipation (outputs not loaded)		594 mW
Input voltage		
Low level		0.7 V max
High level		2.0 V min,Vcc max
Input current (for all inputs)		
Low level	(VIL = 0.4V)	-0.20 mA max
High level	(VIH = 2.4V)	20 uA max
	(VIH = 5.5V)	100 uA max
Output voltage		
Low level	(IOL = 12mA)	0.0V min,0.4 V max
High level	(IOH = -3 mA)	2.4V min,Vcc max
Output drive current maximum capability		
Low level	(VOL = 0.4V)	12.0 mA max
High level	(VOH = 2.4V)	-3.0 mA max
Output leakage current		
Low level	(VOL = 0.4 V)	-20 uA max
High level	(VOH = 2.4 V)	20 uA max

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6.4 Module function. This module consists of four input three-state line drivers. Each driver operates in a single ended driver mode or a high impedance output mode. To operate as 16 single ended line drivers, apply low level to the output disable inputs (to keep the outputs in the normal low impedance mode). All channels operate with no signal inversion. When operating in a bus-oriented system with outputs tied directly to outputs of similar devices, all devices except one shall be placed in the high-impedance state. This is accomplished by insuring that a high level is applied to the output disable inputs of all devices except the one that is used to drive the bus. The driving device provides good capacitive drive capability and waveform integrity especially on the positive transition.

6.4.1 Input to output relationship. Each buffer has the input to output relationship as shown in table II for positive logic.

TABLE II. Input to output relationship.

Inputs		Outputs
Disable	Data	Data out
L	L	L
L	H	H
H	X	High impedance

NOTES: H = High level
L = Low level
X = Don't care

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table III, figure 5.

(b) Continuity test - table IV, figure 6.

(c) Vcc stress test - table V, figure 7.

(d) Ibd test - table VI, figure 8.

(e) Ios test - table VII, figure 9.

(f) Vic test - table VIII, figure 10.

(g) Iozh test - table IX, figure 11.

(h) Iozl test - table X, figure 11.

(i) Vol test - table XI, figures 12 and 13.

(j) Voh test - table XII, figures 12 and 13.

(k) Iih test - table XIII, figure 14.

(l) Iil test - table XIV, figure 14.

(m) Icc test - table XV, figure 15.

(n) tPLH test - table XVI, figures 13, 16 and 17.

(o) tPHL test - table XVII, figures 13, 16 and 17.

(p) tPLZ test - table XVIII, figures 13, 18 and 19.

(q) tPZL test - table XIX, figures 13, 18 and 19.

(r) tPHZ test - table XX, figures 13, 18 and 19.

(s) tPZH test - table XXI, figures 13, 18 and 19.

(t) Xtalkl test - table XXII, figures 13, 20 and 21.

(u) Xtalkd test - table XXIII, figures 13, 20 and 21.

(v) Xtalkd test - table XXIV, figures 13, 20 and 22.

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(w) PSTA test - table XXV, figures 23 and 24.

(x) PSFC test - table XXVI, figure 25.

(y) Cin test - table XXVII, figure 25.

(z) Cout test - table XXVIII, figure 25.

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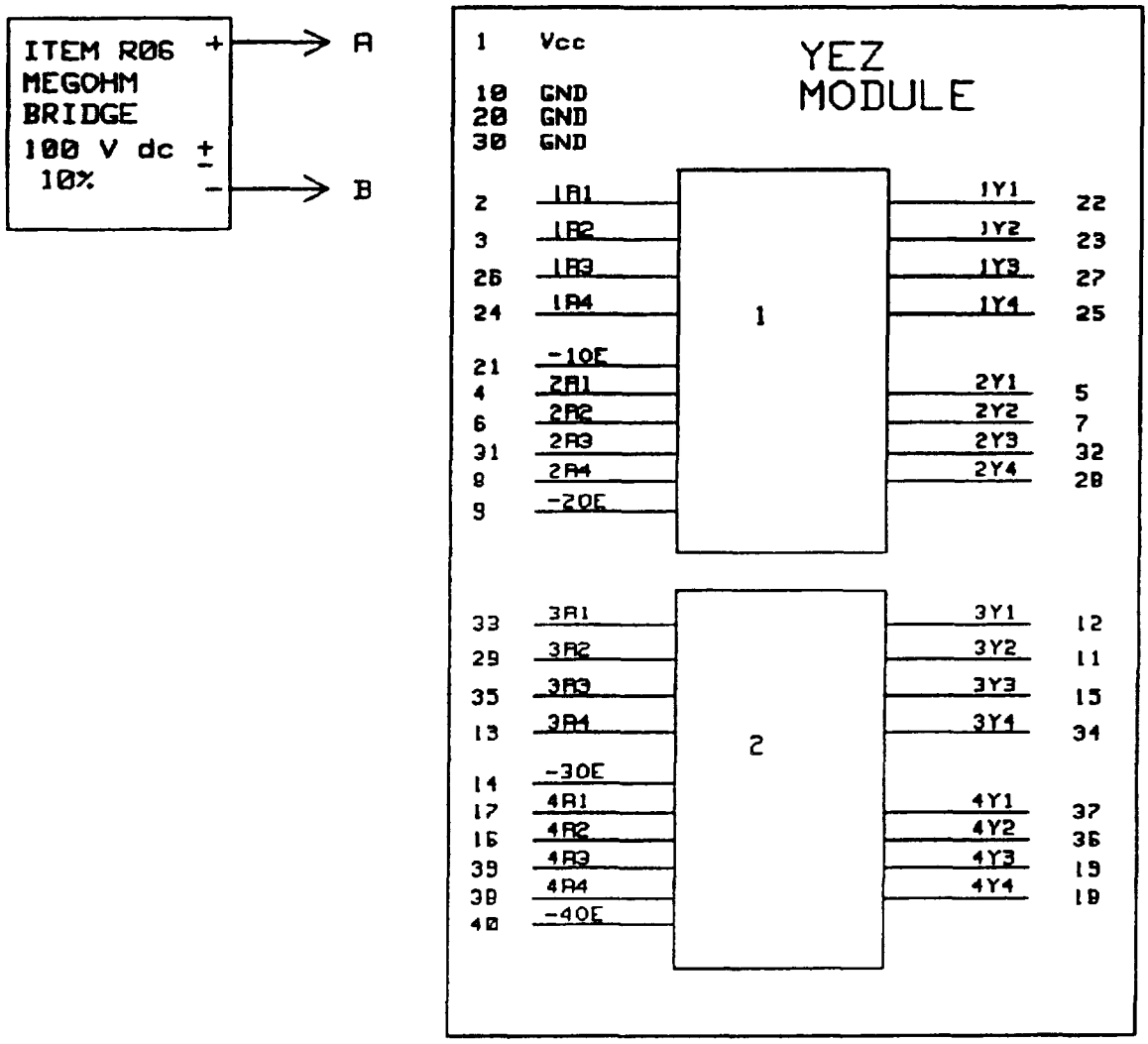
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TABLE III. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
A	Isolation, frame to module pins	All pins	Frame	10	Inf	MOhm	5

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NOTES:

1. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

2. Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

3. Abbreviations used are defined in 6.1.

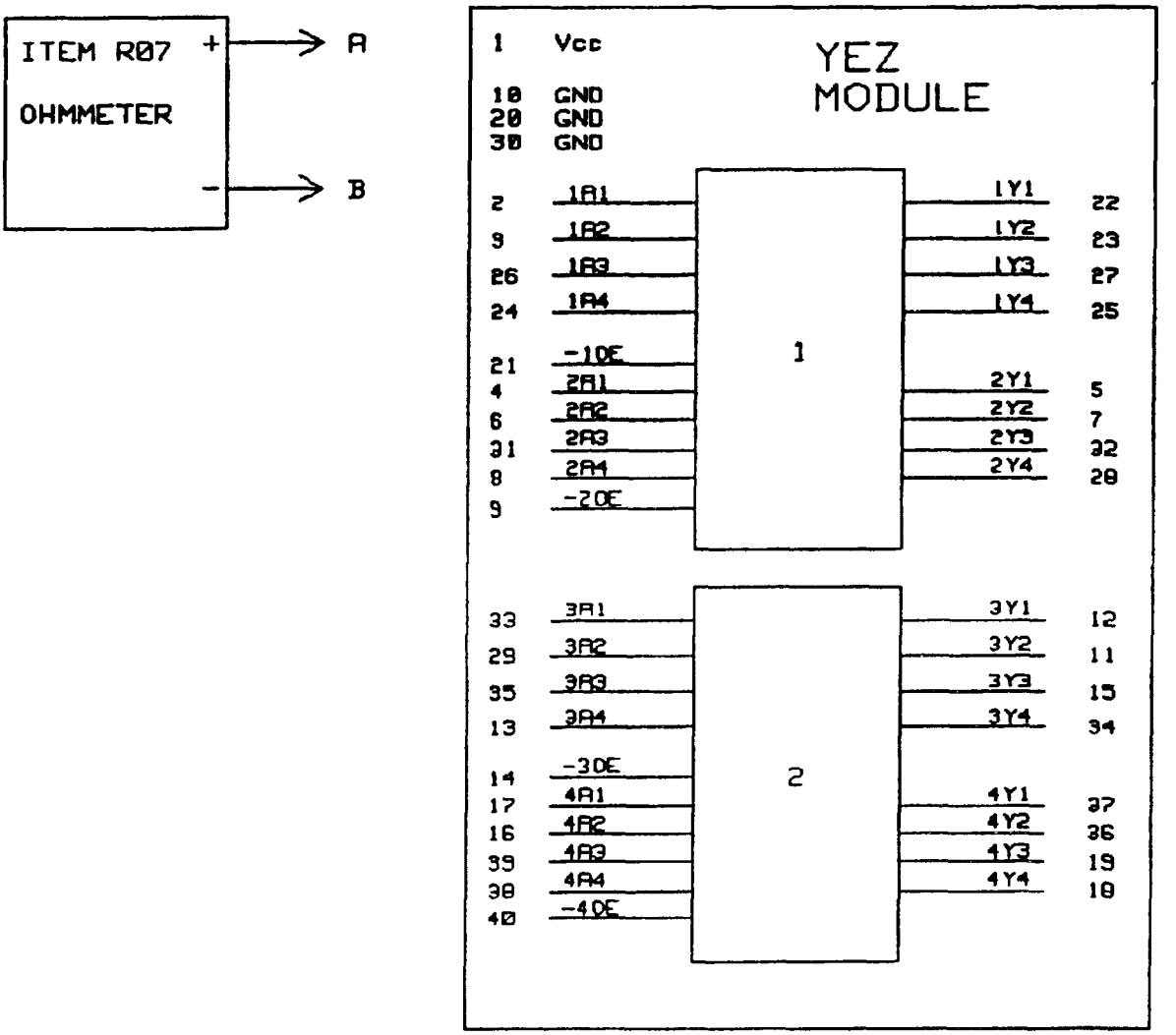
FIGURE 5. YEZ isolation test circuit for table III.

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TABLE IV. Continuity test.

Test cond	Test description	Conct		25 ± 5°C Initial limits		Units	Test fig
		A to pin	B to pin	Min	Max		
B1	Continuity	10	20	0.0	1.0	Ohm	6
B2	Continuity	10	30	0.0	1.0	Ohm	6
B3	Continuity	10	20	0.0	0.1	Ohm	6
B4	Continuity	10	30	0.0	0.1	Ohm	6

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NOTE:

1. Abbreviations used are defined in 6.1.

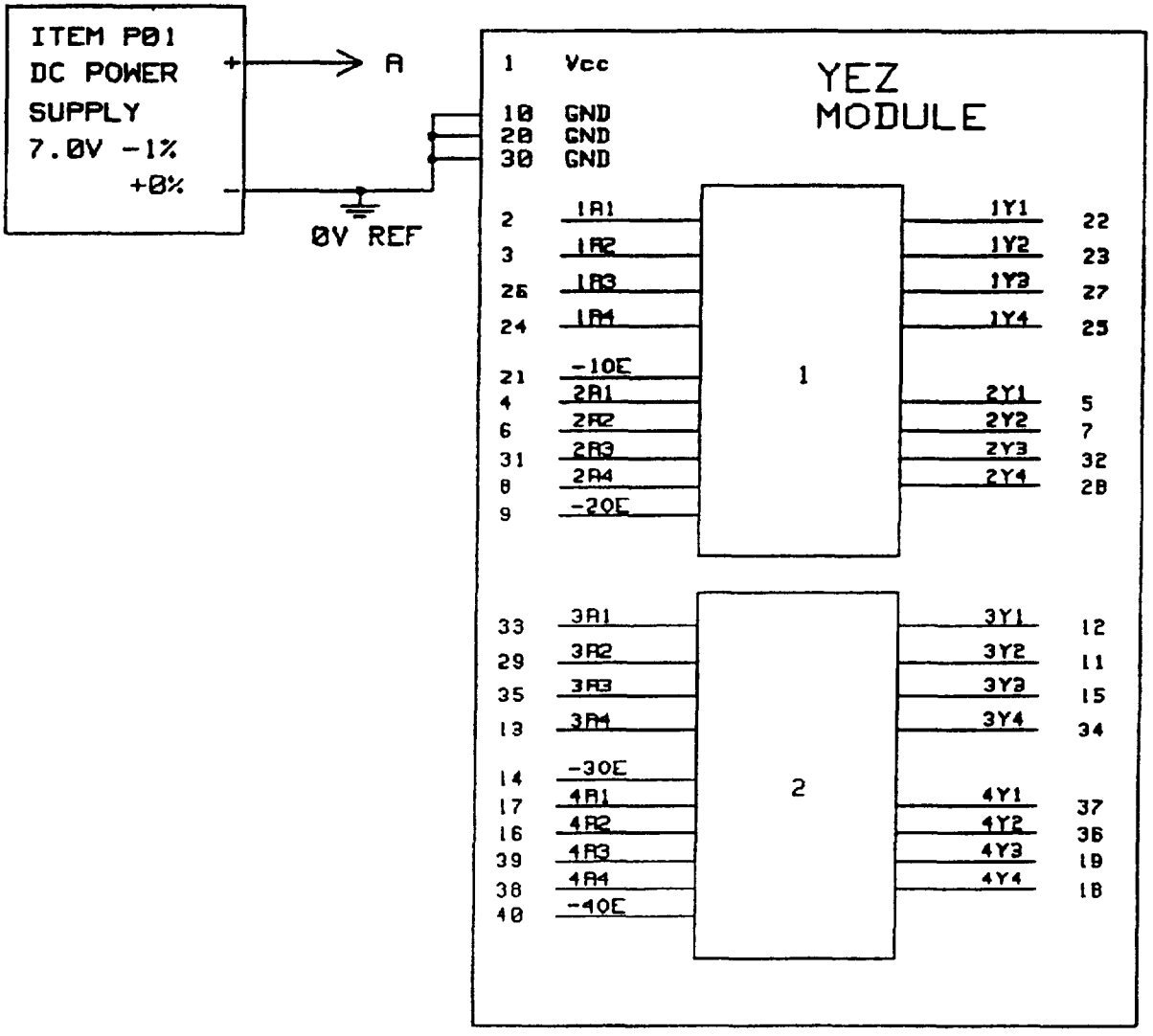
FIGURE 6. Continuity test circuit for table IV.

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TABLE V. Power supply stress test.

Test cond	Parameter	Connect A to pin	25 ± 5°C Initial limits	Units	Test fig
C	Vcc stress	1	Module shall pass all remaining tests.	NA	7

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NOTES:

1. Vcc shall be connected for a minimum of 5 seconds.
2. When performing this test, no measurements shall be taken.
3. Abbreviations used are defined in 6.1.

FIGURE 7. YEZ Vcc stress test circuit for table V.

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TABLE VI. Input breakdown current test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
D1	1A1	3,4,6,8,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40	2	0	100	uA	8
D2	1A2	2,4,6,8,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40	3				
D3	1A3	2,3,4,6,8,9,13, 14,16,17,21,24, 29,31,33,35,38, 39,40	26				
D4	1A4	2,3,4,6,8,9,13, 14,16,17,21,26, 29,31,33,35,38, 39,40	24				
D5	2A1	2,3,6,8,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40	4				
D6	2A2	2,3,4,8,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40	6				
D7	2A3	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,33,35,38, 39,40	31				
D8	2A4	2,3,4,6,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40	8	0	100	uA	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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TABLE VI. Input breakdown current test - Continued.

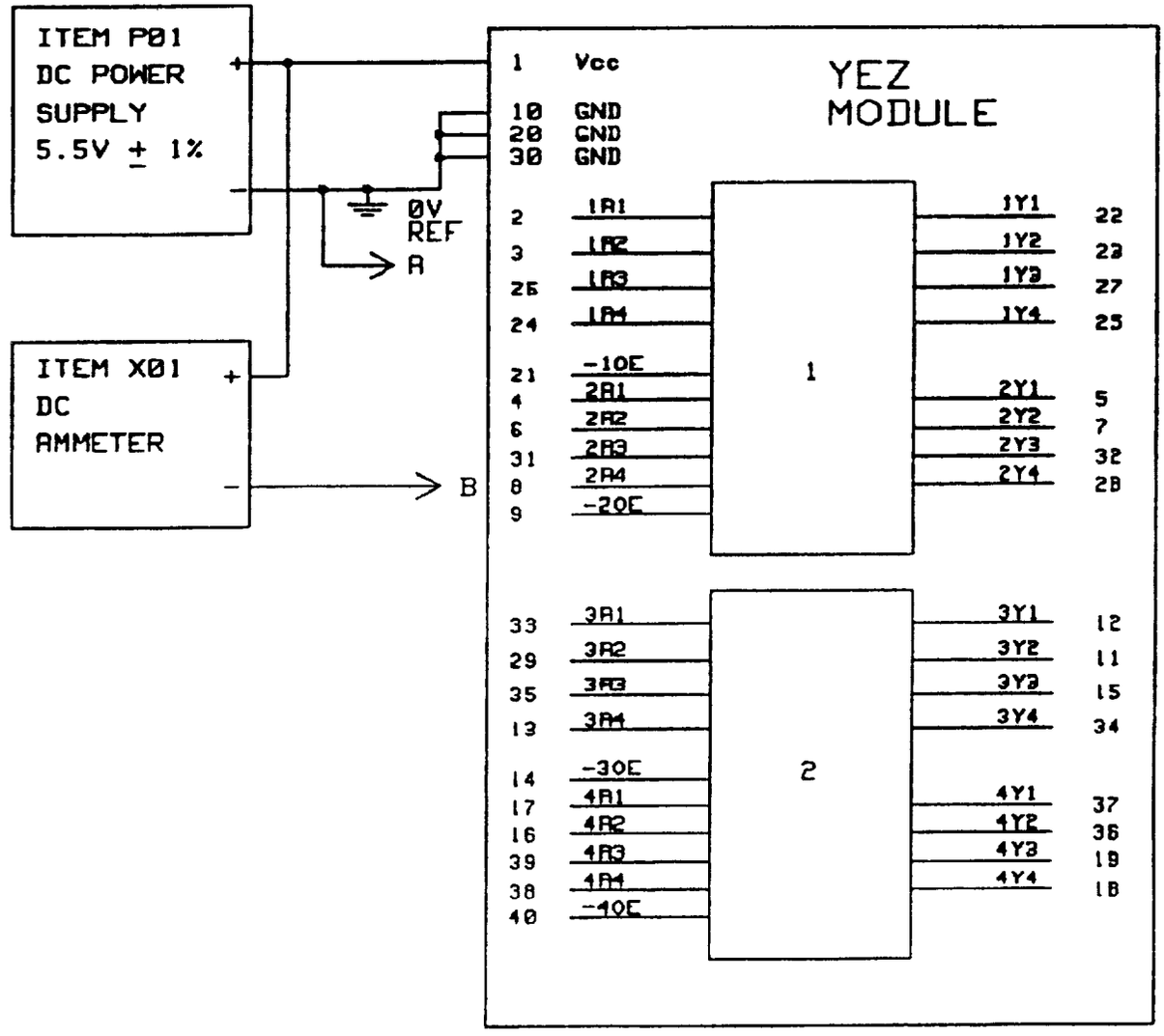
Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
D9	3A1	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,35,38, 39,40	33	0	100	uA	8
D10	3A2	2,3,4,6,8,9,13, 14,16,17,21,24, 26,31,33,35,38, 39,40	29				
D11	3A3	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,33,38, 39,40	35				
D12	3A4	2,3,4,6,8,9,13, 14,16,21,24,26, 29,31,33,35,38, 39,40	13				
D13	4A1	2,3,4,6,8,9,13, 14,16,21,24,26, 29,31,33,35,38, 39,40	17				
D14	4A2	2,3,4,6,8,9,13, 14,17,21,24,26, 29,31,33,35,38, 39,40	16				
D15	4A3	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,33,35, 38,40	39				
D16	4A4	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,33,35, 39,40	38	0	100	uA	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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TABLE VI. Input breakdown current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
D17	-10E	2,3,4,6,8,9,13, 14,16,17,24,26, 29,33,35,38,39, 40,31	21	0	100	uA	8
D18	-20E	2,3,4,6,8,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40	9				
D19	-30E	2,3,4,6,8,9,13, 16,17,21,24,26, 29,31,33,35,38, 39,40	14				
D20	-40E	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,33,35, 38,39	40	0	100	uA	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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NOTES:

1. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

2. Abbreviations used are defined in 6.1.

FIGURE 8. Input breakdown current test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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TABLE VII. Output short circuit current test.

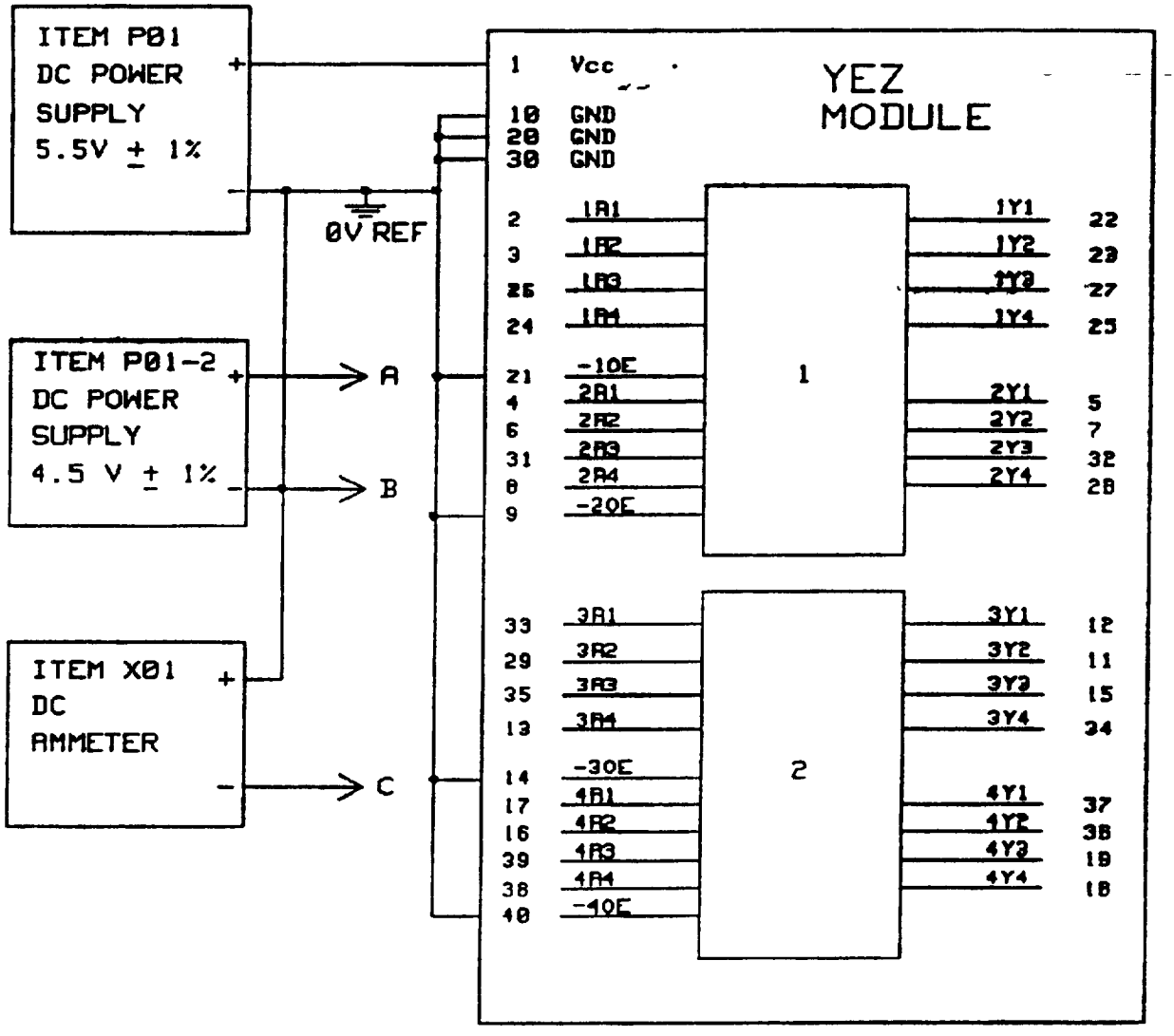
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
E1	1Y1	2	3,4,6,8,13,16, 17,24,26,29,31 33,35,38,39	22	-225	-40	mA	9
E2	1Y2	3	2,4,6,8,13,16, 17,24,26,29,31 33,35,38,39	23				
E3	1Y3	26	2,3,4,6,8,13, 16,17,24,29,31 33,35,38,39	27				
E4	1Y4	24	2,3,4,6,8,13, 16,17,26,29,31 33,35,38,39	25				
E5	2Y1	4	2,3,6,8,13,16, 17,24,26,29,31 33,35,38,39	5				
E6	2Y2	6	2,3,4,8,13,16, 17,24,26,29,31 33,35,38,39	7				
E7	2Y3	31	2,3,4,6,8,13, 16,17,24,26,29 33,35,38,39	32				
E8	2Y4	8	2,3,4,6,13,16, 17,24,26,29,31 33,35,38,39	28				
E9	3Y1	33	2,3,4,6,8,13, 16,17,24,26,29 31,35,38,39	12				
E10	3Y2	29	2,3,4,6,8,13, 16,17,24,26,31 33,35,38,39	11	-225	-40	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
E11	3Y3	35	2,3,4,6,8,13, 16,17,24,26,29 31,33,38,39	15	-225	-40	mA	9
E12	3Y4	13	2,3,4,6,8,16, 17,24,26,29,31 33,35,38,39	34				
E13	4Y1	17	2,3,4,6,8,13, 16,24,26,29,31 33,35,38,39	37				
E14	4Y2	16	2,3,4,6,8,13, 17,24,26,29,31 33,35,38,39	36				
E15	4Y3	39	2,3,4,6,8,13, 16,17,24,26,29 31,33,35,38	19				
E16	4Y4	38	2,3,4,6,8,13, 16,17,24,26,29 31,33,35,39	18	-225	-40	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
Check the source to verify that this is the current version before use.



NOTES:

1. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

2. Only one output shall be shorted at a time and the measurement taken within 1 second.

3. Abbreviations used are defined in 6.1.

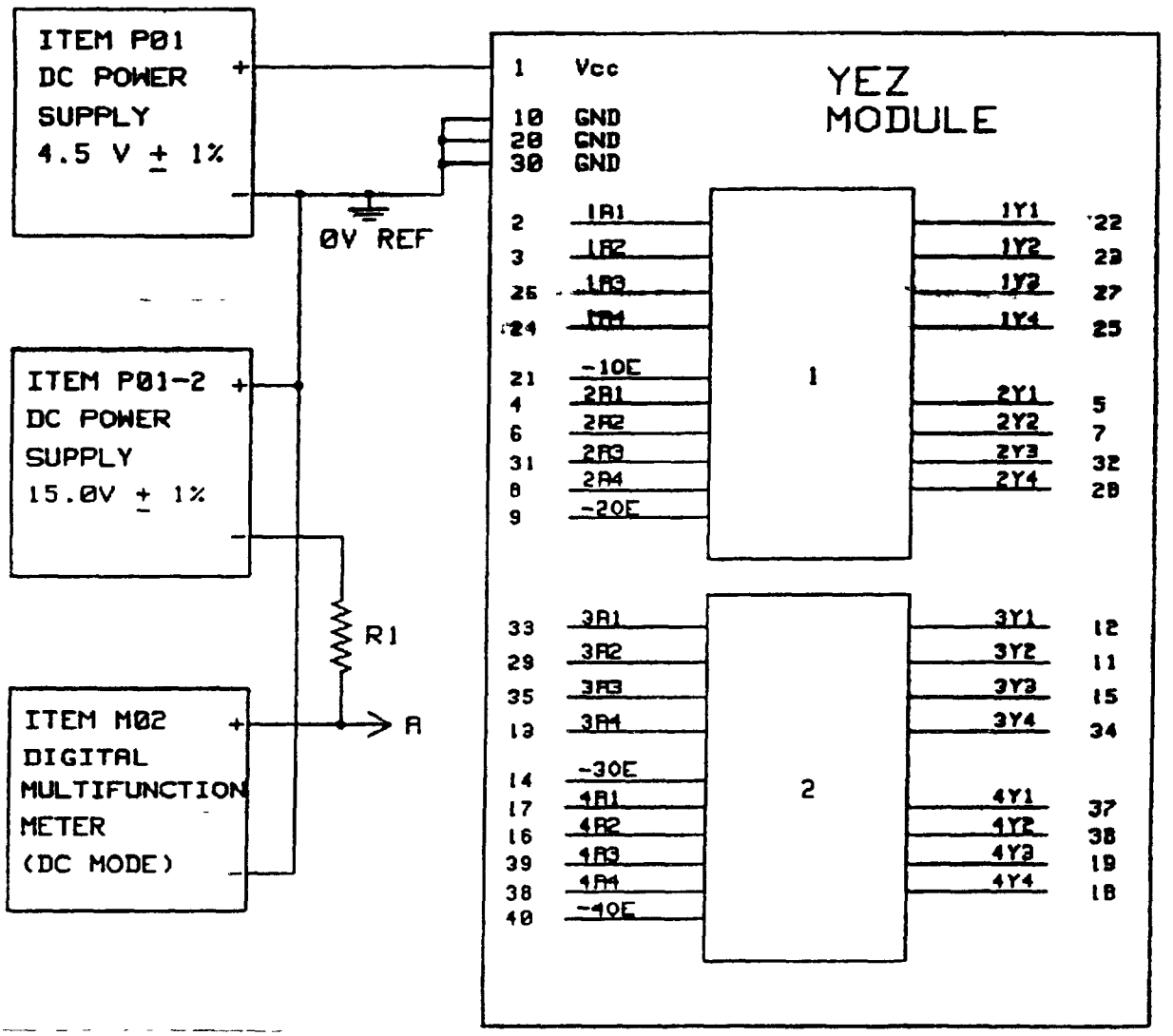
FIGURE 9. Output short circuit current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test cond	Signal name	Connect A to Pin	25 ± 5°C Initial limits Min	Max	Unit	Test fig
F1	1A1	2	-1.5	-0.4	V	10
F2	1A2	3				
F3	1A3	26				
F4	1A4	24				
F5	2A1	4				
F6	2A2	6				
F7	2A3	31				
F8	2A4	8				
F9	3A1	33				
F10	3A2	29				
F11	3A3	35				
F12	3A4	13				
F13	4A1	17				
F14	4A2	16				
F15	4A3	39				
F16	4A4	38				
F17	-10E	21				
F18	-20E	9				
F19	-30E	14				
F20	-40E	40	-1.5	-0.4	V	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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NOTES:

1. R1 = 750 Ohm ± 1 percent, 1/4 W.
2. Abbreviations used are defined in 6.1.

FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Output leakage current, high level, test.

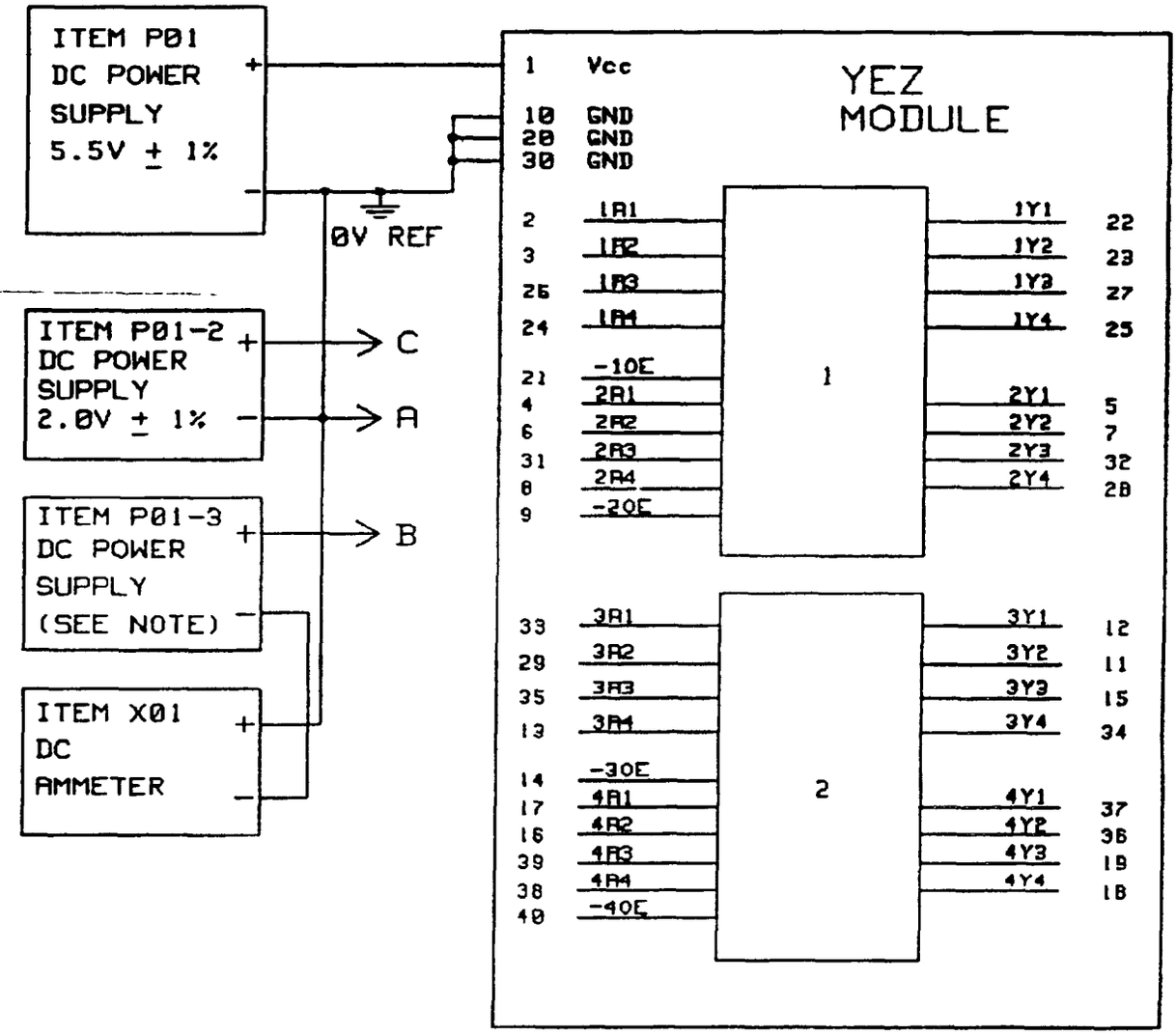
Test cond	Signal name	Connect			25 ± 5°C Initial limits		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
G1	1Y1	Open	22	2,21	-20	20	uA	11
G2	1Y2		23	3,21				
G3	1Y3		27	26,21				
G4	1Y4		25	24,21				
G5	2Y1		5	4,9				
G6	2Y2		7	6,9				
G7	2Y3		32	31,9				
G8	2Y4		28	8,9				
G9	3Y1		12	33,14				
G10	3Y2		11	29,14				
G11	3Y3		15	35,14				
G12	3Y4		34	13,14				
G13	4Y1		37	17,40				
G14	4Y2		36	16,40				
G15	4Y3		19	39,40				
G16	4Y4	Open	18	38,40	-20	20	uA	11

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TABLE X. Output leakage current, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
H1	1Y1	2	22	21	-20	20	uA	11
H2	1Y2	3	23	21				
H3	1Y3	26	27	21				
H4	1Y4	24	25	21				
H5	2Y1	4	5	9				
H6	2Y2	6	7	9				
H7	2Y3	31	32	9				
H8	2Y4	8	28	9				
H9	3Y1	33	12	14				
H10	3Y2	29	11	14				
H11	3Y3	35	15	14				
H12	3Y4	13	34	14				
H13	4Y1	17	37	40				
H14	4Y2	16	36	40				
H15	4Y3	39	19	40				
H16	4Y4	38	18	40	-20	20	uA	11

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NOTES:

1. Caution: Do not isolate the chassis of X01 ammeter from earth ground.

2. For test condition:
   G1-G16 - Item P01-3 = 2.7 V ± 1 percent.
   H1-H16 - Item P01-3 = 0.4 V ± 1 percent.

3. Abbreviations used are defined in 6.1.

FIGURE 11. Output leakage current, high and low level, test circuit for tables IX and X.

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TABLE XI. Output voltage, low level, test.

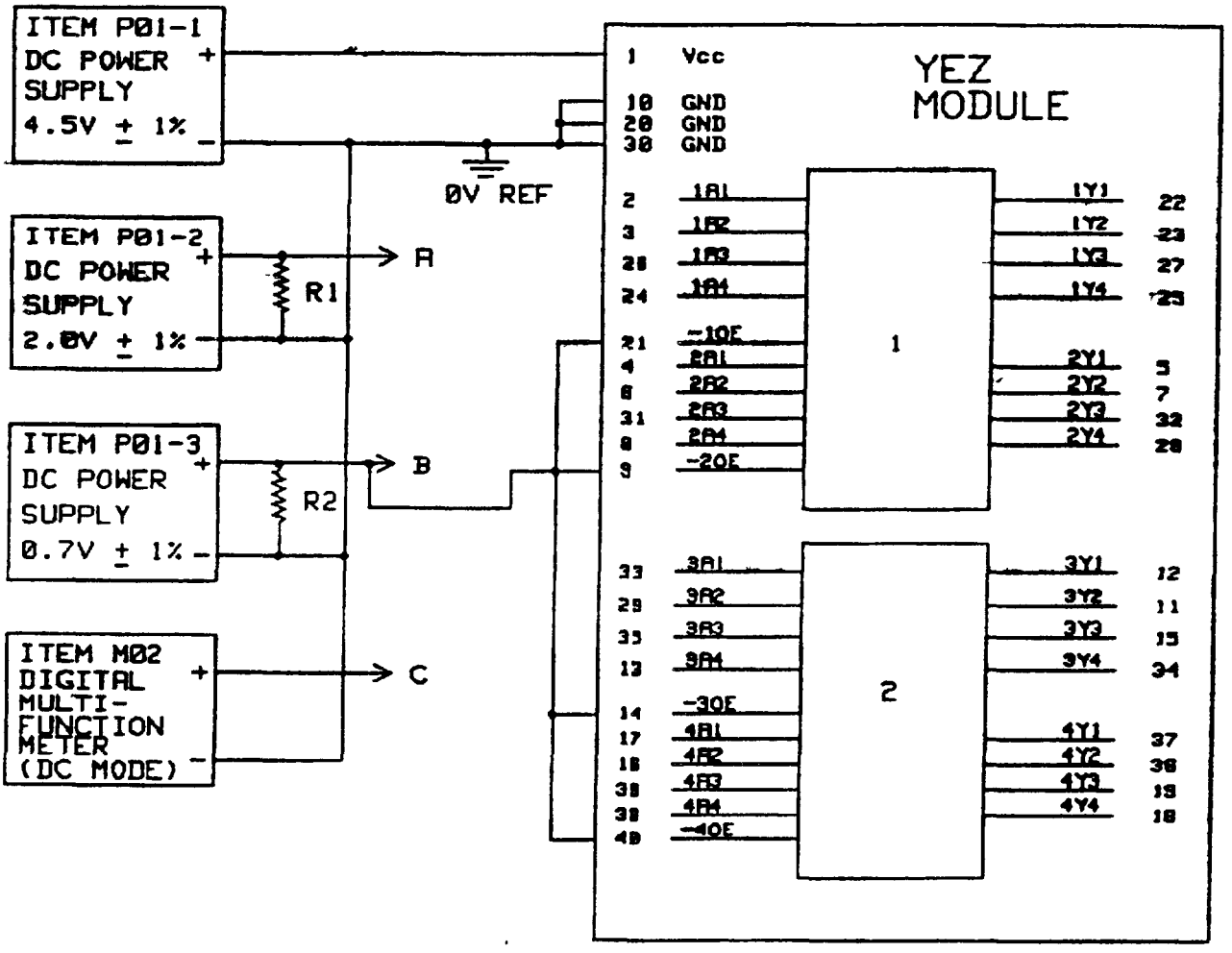
Test cond	Signal name	Connect			25 ± 5°C Initial limits		Unit	Test figs
		A to pin	B to pin	C to pin	Min	Max		
J1	1Y1	3,26,24	2	22	0.0	0.4	V	12, 13
J2	1Y2	2,26,24	3	23				
J3	1Y3	2,3,24	26	27				
J4	1Y4	2,3,26	24	25				
J5	2Y1	6,31,8	4	5				
J6	2Y2	4,31,8	6	7				
J7	2Y3	4,6,8	31	32				
J8	2Y4	4,6,31	8	28				
J9	3Y1	29,35,13	33	12				
J10	3Y2	33,35,13	29	11				
J11	3Y3	33,29,13	35	15				
J12	3Y4	33,29,35	13	34				
J13	4Y1	16,39,38	17	37				
J14	4Y2	17,39,38	16	36				
J15	4Y3	17,16,38	39	19				
J16	4Y4	17,16,39	38	18	0.0	0.4	V	12, 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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TABLE XII. Output voltage, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
K1	1Y1	2	3,26,24	22	2.4	4.5	V	12, 13
K2	1Y2	3	2,26,24	23				
K3	1Y3	26	2,3,24	27				
K4	1Y4	24	2,3,26	25				
K5	2Y1	4	6,31,8	5				
K6	2Y2	6	4,31,8	7				
K7	2Y3	31	4,6,8	32				
K8	2Y4	8	4,6,31	28				
K9	3Y1	33	29,35,13	12				
K10	3Y2	29	33,35,13	11				
K11	3Y3	35	33,29,13	15				
K12	3Y4	13	33,29,35	34				
K13	4Y1	17	16,39,38	37				
K14	4Y2	16	17,39,38	36				
K15	4Y3	39	17,16,38	19				
K16	4Y4	38	17,16,39	18	2.4	4.5	V	12, 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
Check the source to verify that this is the current version before use.

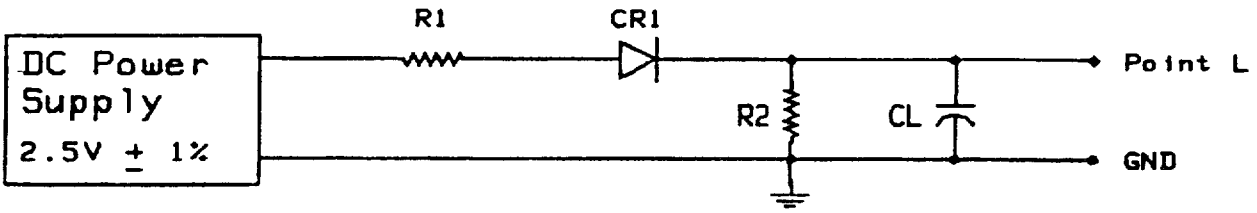


NOTES:

1. R1 = 51.1 Ohms ± 5 percent, 1/2 W.
2. R2 = 51.1 Ohms ± 5 percent, 1/2 W.
3. Load output under test with load as specified on figure 13.
4. Abbreviations used are defined in 6.1.

FIGURE 12. Output voltage, low and high level, test circuit for tables XI and XII.

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NOTES:

1. R2 = 806 ± 1 percent, 1/8W.

2. R1 = select value for current flow of 12.0 mA ± 5 percent out of point L with point L held at 0.4V.

3. CR1 = 1N4150 or equivalent.

4. CL = 150 pF (including probe and parasitic capacitance) for test conditions P1-P16, Q1-Q16, S, and U.

5. CL = 300 pF (including probe and parasitic capacitance) for test conditions P17-P32 and Q17-Q32.

6. CL = 50 pF (including probe and parasitic capacitance) for test conditions R, T, V, W, and X.

7. Capacitor CL is not required for DC electrical test conditions J and K.

8. The lead length from point L to the appropriate module pin shall be less than 1 inch (25.40 mm).

FIGURE 13. YEZ logic load circuit.

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TABLE XIII. Input current, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
L1	1A1	2	3,4,6,8,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40	0	20	uA	14
L2	1A2	3	2,4,6,8,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40				
L3	1A3	26	2,3,4,6,8,9,13, 14,16,17,21,24, 29,31,33,35,38, 39,40				
L4	1A4	24	2,3,4,6,8,9,13, 14,16,17,21,26, 29,31,33,35,38, 39,40				
L5	2A1	4	2,3,6,8,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40				
L6	2A2	6	2,3,4,8,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40				
L7	2A3	31	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,33,35,38, 39,40				
L8	2A4	8	2,3,4,6,9,13,14, 16,17,21,24,26, 29,31,33,35,38, 39,40	0	20	uA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current, high level, test - Continued.

Test cond	Signal name	Connect		25 ± 5°C Initial limits		Unit	Test fig
		A to pin	B to pin	Min	Max		
L9	3A1	33	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,35,38, 39,40	0	20	uA	14
L10	3A2	29	2,3,4,6,8,9,13, 14,16,17,21,24, 26,31,33,35,38,				
L11	3A3	35	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,33,38, 39,40				
L12	3A4	13	2,3,4,6,8,9,14, 16,17,21,24,26, 29,31,33,35,38, 39,40				
L13	4A1	17	2,3,4,6,8,9,13, 14,16,21,24,26, 29,31,33,35,38, 39,40				
L14	4A2	16	2,3,4,6,8,9,13, 14,17,21,24,26, 29,31,33,35,38, 39,40				
L15	4A3	39	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,33,35, 38,40				
L16	4A4	38	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,33,35, 39,40	0	20	uA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current, high level, test - Continued.

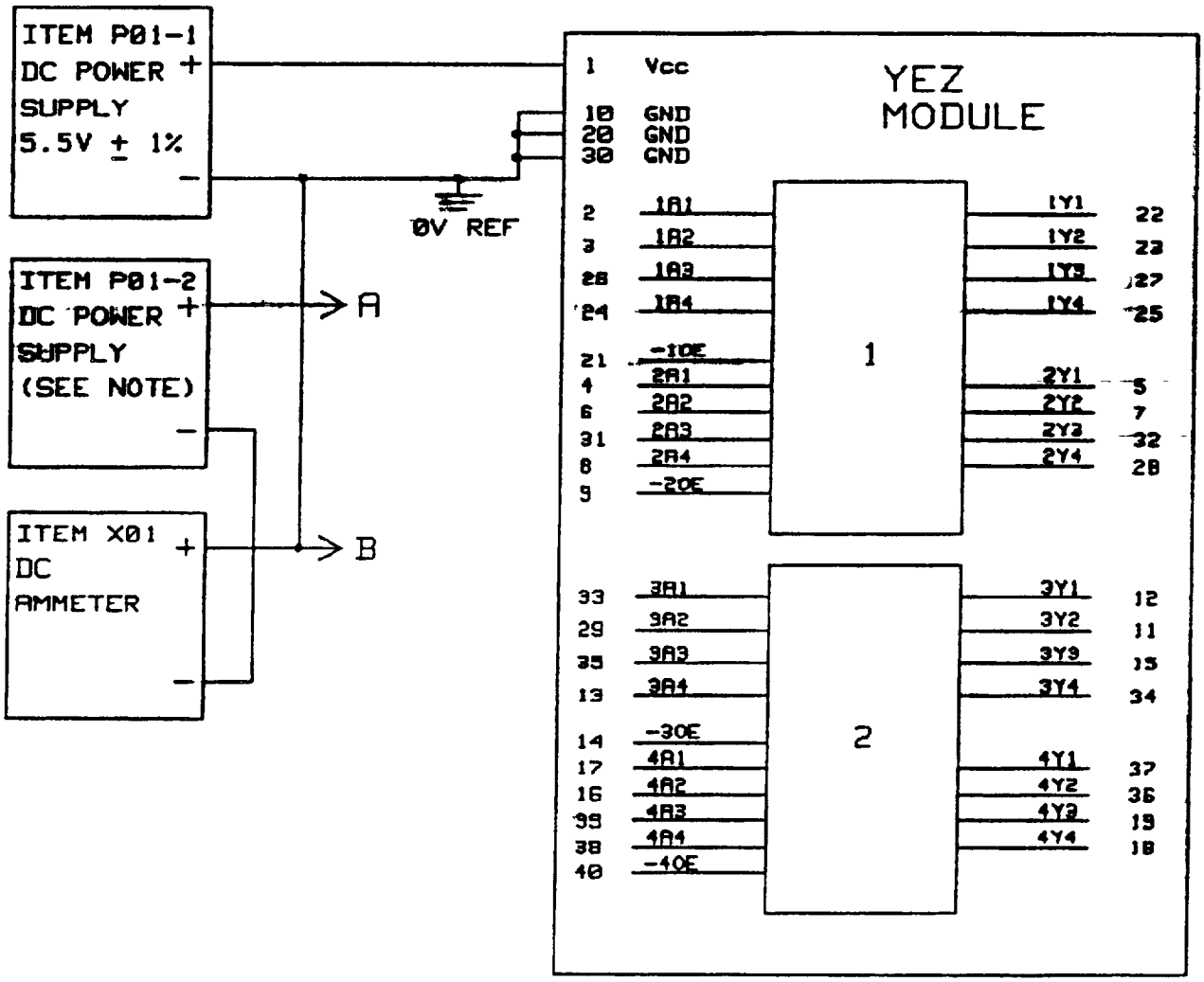
Test cond	Signal name	Connect		25 ± 5°C Initial limits		Unit	Test fig
		A to pin	B to pin	Min	Max		
L17	-10E	21	2,3,4,6,8,9,13, 14,16,17,24,26, 29,31,33,35,38, 39,40	0	20	uA	14
L18	-20E	9	2,3,4,6,8,13,14 16,17,21,24,26, 29,31,33,35,38, 39,40				
L19	-30E	14	2,3,4,6,8,9,13, 16,17,21,24,26, 29,31,33,35,38, 39,40				
L20	-40E	40	2,3,4,6,8,9,13, 14,16,17,21,24, 26,29,31,33,35, 38,39	0	20	uA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
Check the source to verify that this is the current version before use.

TABLE XIV. Input current, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test fig
M1	1A1	2	Open	-0.2	0.0	mA	14
M2	1A2	3					
M3	1A3	26					
M4	1A4	24					
M5	2A1	4					
M6	2A2	6					
M7	2A3	31					
M8	2A4	8					
M9	3A1	33					
M10	3A2	29					
M11	3A3	35					
M12	3A4	13					
M13	4A1	17					
M14	4A2	16					
M15	4A3	39					
M16	4A4	38					
M17	-10E	21					
M18	-20E	9					
M19	-30E	14					
M20	-40E	40	Open	-0.2	0.0	mA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
Check the source to verify that this is the current version before use.



NOTES:

1. Caution: Do not isolate the chassis of X01 from earth ground.

2. For test conditions:
   L1-L20 - Item P01-2 = 2.7V ± 1 percent.
   M1-M20 - Item P01-2 = 0.4V ± 1 percent.

3. Abbreviations used are defined in 6.1.

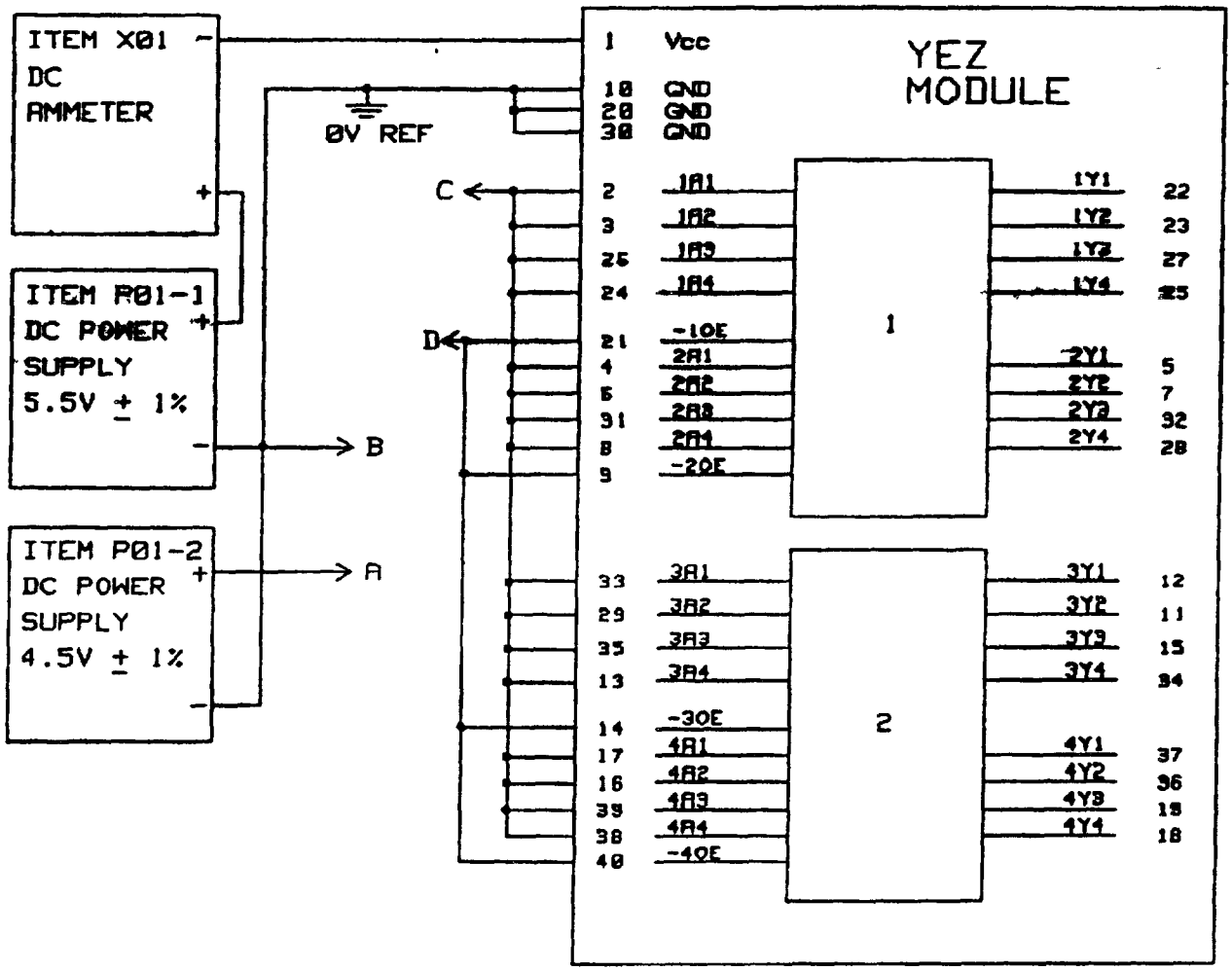
FIGURE 14. Input current, high and low level, test circuit for tables XIII and XIV.

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TABLE XV. Power supply current, high and low level and high impedance, test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Unit	Test fig
		A to pin	B to pin	Min	Max		
N1	Power supply current, high level	C	D	5.4	54	mA	15
N2	Power supply current, low level	Open	C,D	9.2	92	mA	15
N3	Power supply current, high impedance	D	Open	10.8	108	mA	15

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NOTE:

1. Caution: Do not isolate the chassis of X01 ammeter from earth ground.

2. Abbreviations used are defined in 6.1.

FIGURE 15. Power supply current, high and low level and high impedance, test circuit for table XV.

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TABLE XVI. Propagation delay time (tPLH) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
P1	1Y1	2	22	5	29	nS	16, 17, 13
P2	1Y2	3	23				
P3	1Y3	26	27				
P4	1Y4	24	25				
P5	2Y1	4	5				
P6	2Y2	6	7				
P7	2Y3	31	32				
P8	2Y4	8	28				
P9	3Y1	33	12				
P10	3Y2	29	11				
P11	3Y3	35	15				
P12	3Y4	13	34				
P13	4Y1	17	37				
P14	4Y2	16	36				
P15	4Y3	39	19				
P16	4Y4	38	18	5	29	nS	
P17	1Y1	2	22	10	38	nS	16, 17, 13,
P18	1Y2	3	23	10	38	nS	

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TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
P19	1Y3	26	27	10	38	nS	16, 17, 13
P20	1Y4	24	25				
P21	2Y1	4	5				
P22	2Y2	6	7				
P23	2Y3	31	32				
P24	2Y4	8	28				
P25	3Y1	33	12				
P26	3Y2	29	11				
P27	3Y3	35	15				
P28	3Y4	13	34				
P29	4Y1	17	37				
P30	4Y2	16	36				
P31	4Y3	39	19				16, 17,
P32	4Y4	38	18	10	38	nS	13

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TABLE XVII. Propagation delay time (tPHL) test.

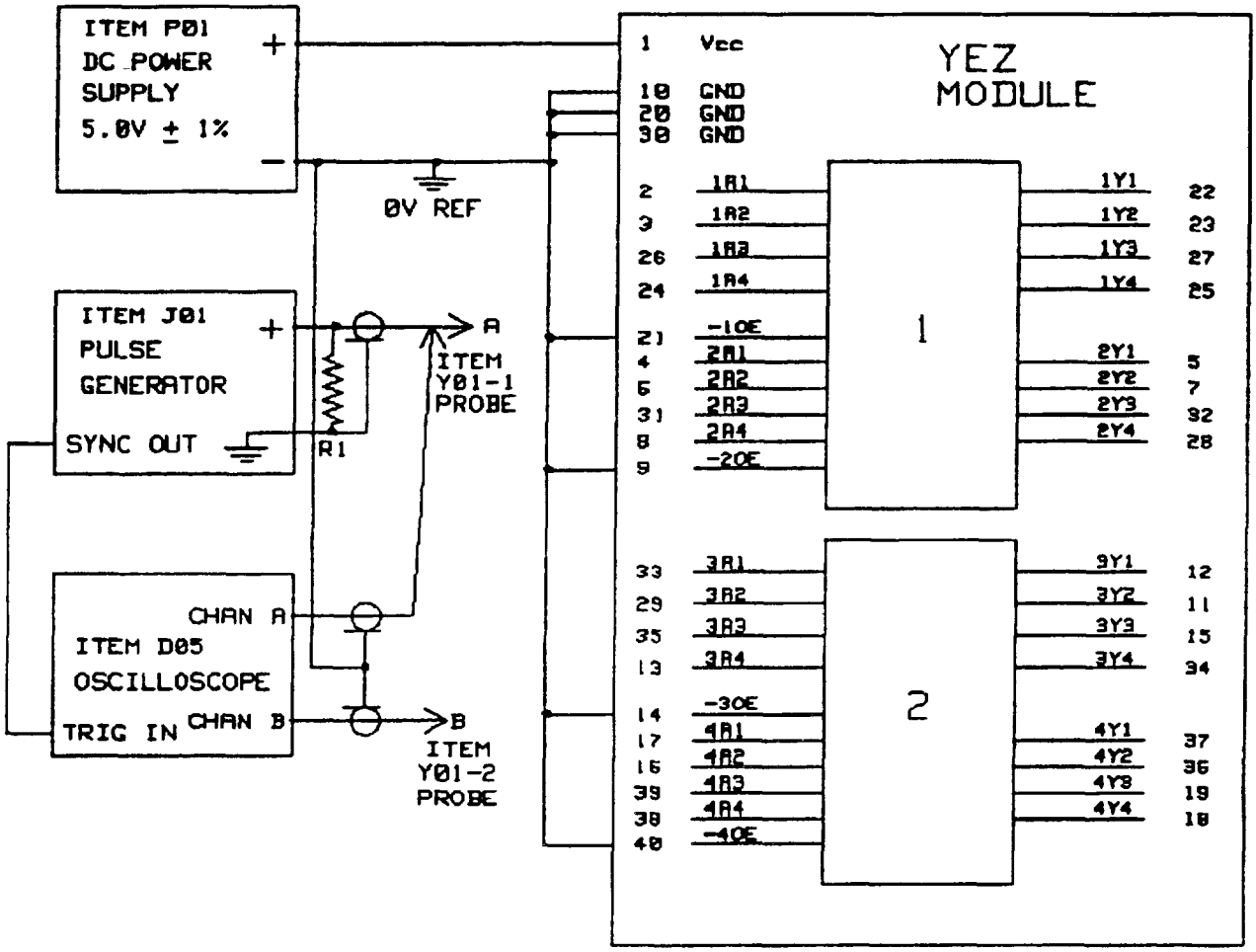
Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
Q1	1Y1	2	22	5	26	nS	16, 17, 13
Q2	1Y2	3	23				
Q3	1Y3	26	27				
Q4	1Y4	24	25				
Q5	2Y1	4	5				
Q6	2Y2	6	7				
Q7	2Y3	31	32				
Q8	2Y4	8	28				
Q9	3Y1	33	12				
Q10	3Y2	29	11				
Q11	3Y3	35	15				
Q12	3Y4	13	34				
Q13	4Y1	17	37				
Q14	4Y2	16	36				
Q15	4Y3	39	19				
Q16	4Y4	38	18	5	26		
Q17	1Y1	2	22	10	32		16, 17, 13
Q18	1Y2	3	23	10	32	nS	

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TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
Q19	1Y3	26	27	10	32	nS	16, 17, 13
Q20	1Y4	24	25				
Q21	2Y1	4	5				
Q22	2Y2	6	7				
Q23	2Y3	31	32				
Q24	2Y4	8	28				
Q25	3Y1	33	12				
Q26	3Y2	29	11				
Q27	3Y3	35	15				
Q28	3Y4	13	34				
Q29	4Y1	17	37				
Q30	4Y2	16	36				
Q31	4Y3	39	19				
Q32	4Y4	38	18	10	32	nS	16, 17, 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
Check the source to verify that this is the current version before use.

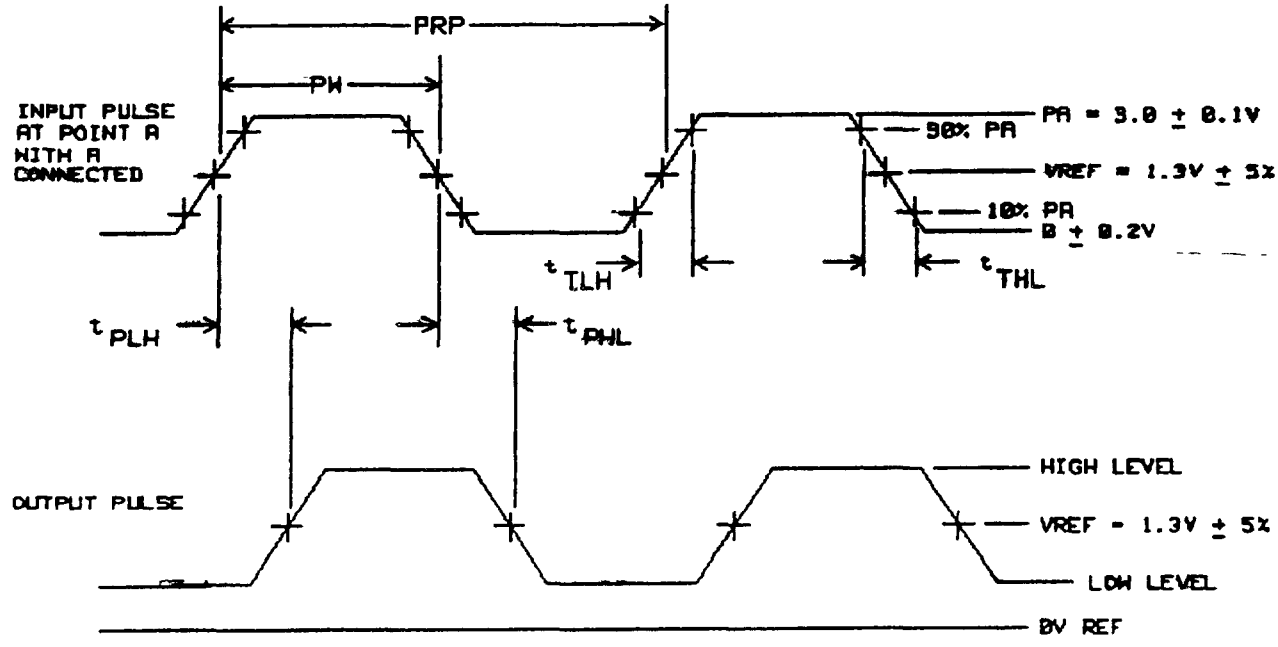


NOTES:

1. R1 = 51.1 Ohms ± 5 percent, 1/2W.
2. Load output under test with load as specified on figure 13.
3. Abbreviations used are defined in 6.1.

FIGURE 16. Propagation delay time (tPLH and tPHL) test circuit for tables XVI and XVII.

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NOTES:

1. Input pulse parameters:
   PW = 500 ± 50 nS
   PRP = 1000 ± 100 nS
   tTLH ≤ 15 nS, tTHL ≤ 6 nS.

2. Before making the measurements, the output levels shall be verified to be ≤ 0.4V for the low level and ≥ 2.4V for the high level.

3. All voltages shall be referenced to ground.

FIGURE 17. Propagation delay time (tPLH and tPHL) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Disable delay time (tPLZ) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
R1	1Y1	Open	2,3,26, 24	21	22	5	30	nS	18, 19, 13
R2	1Y2				23				
R3	1Y3				27				
R4	1Y4		2,3,26, 24	21	25				
R5	2Y1		31,4,6,8	9	5				
R6	2Y2			9	7				
R7	2Y3			9	32				
R8	2Y4		31,4,6,8	9	28				
R9	3Y1		13,29, 33,35	14	12				
R10	3Y2			14	11				
R11	3Y3			14	15				
R12	3Y4		13,29, 33,35	14	34				
R13	4Y1		16,17, 38,39	40	37				
R14	4Y2			40	36				
R15	4Y3			40	19				
R16	4Y4	Open	16,17, 38,39	40	18	5	30	nS	18, 19, 13

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TABLE XIX. Enable delay time (tPZL) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
S1	1Y1	Open	2,3,26, 24	21	22	5	40	nS	18, 19, 13
S2	1Y2				23				
S3	1Y3				27				
S4	1Y4		2,3,26, 24	21	25				
S5	2Y1		31,4,6,8	9	5				
S6	2Y2			9	7				
S7	2Y3			9	32				
S8	2Y4		31,4,6,8	9	28				
S9	3Y1		13,29, 33,35	14	12				
S10	3Y2			14	11				
S11	3Y3			14	15				
S12	3Y4		13,29, 33,35	14	34				
S13	4Y1		16,17, 38,39	40	37				
S14	4Y2			40	36				
S15	4Y3			40	19				
S16	4Y4	Open	16,17, 38,39	40	18	5	40	nS	18, 19, 13

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TABLE XX. Disable delay time (tPHZ) test.

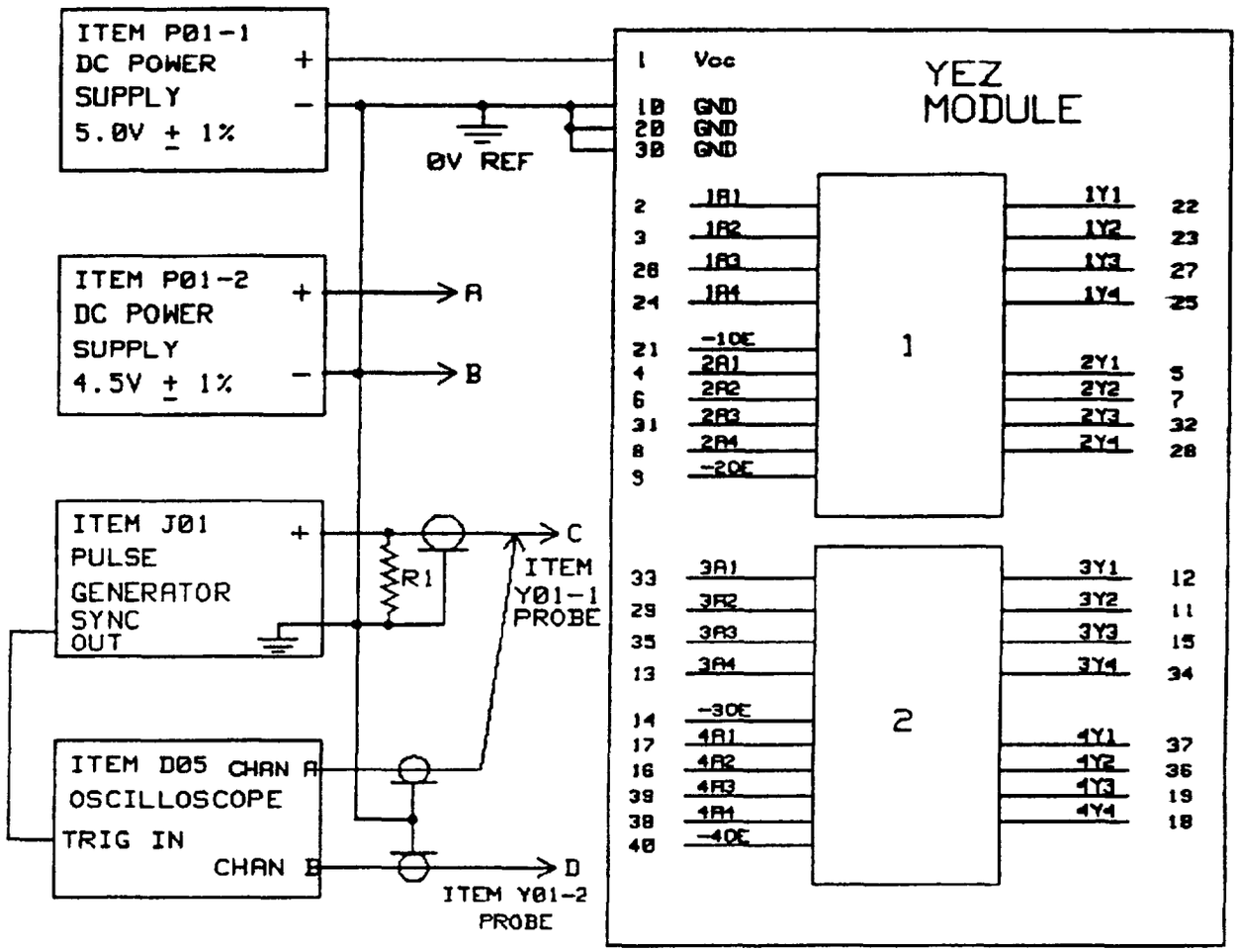
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
T1	1Y1	2,3,26, 24	Open	21	22	5	35	nS	18, 19, 13
T2	1Y2				23				
T3	1Y3				27				
T4	1Y4	2,3,26, 24		21	25				
T5	2Y1	31,4,6,8		9	5				
T6	2Y2			9	7				
T7	2Y3			9	32				
T8	2Y4	31,4,6,8		9	28				
T9	3Y1	13,29, 33,35		14	12				
T10	3Y2			14	11				
T11	3Y3			14	15				
T12	3Y4	13,29, 33,35		14	34				
T13	4Y1	16,17, 38,39		40	37				
T14	4Y2			40	36				
T15	4Y3			40	19				
T16	4Y4	16,17, 38,39	Open	40	18	5	35	nS	18, 19, 13

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TABLE XXI. Enable delay time (tPZH) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
U1	1Y1	2,3,26, 24	Open	21	22	5	40	nS	18, 19,
U2	1Y2				23				13
U3	1Y3				27				
U4	1Y4	2,3,26, 24		21	25				
U5	2Y1	31,4,6,8		9	5				
U6	2Y2			9	7				
U7	2Y3			9	32				
U8	2Y4	31,4,6,8		9	28				
U9	3Y1	13,29, 33,35		14	12				
U10	3Y2			14	11				
U11	3Y3			14	15				
U12	3Y4	13,29, 33,35		14	34				
U13	4Y1	16,17, 38,39		40	37				
U14	4Y2			40	36				
U15	4Y3			40	19				18,
U16	4Y4	16,17, 38,39	Open	40	18	5	40	nS	19, 13

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NOTES:

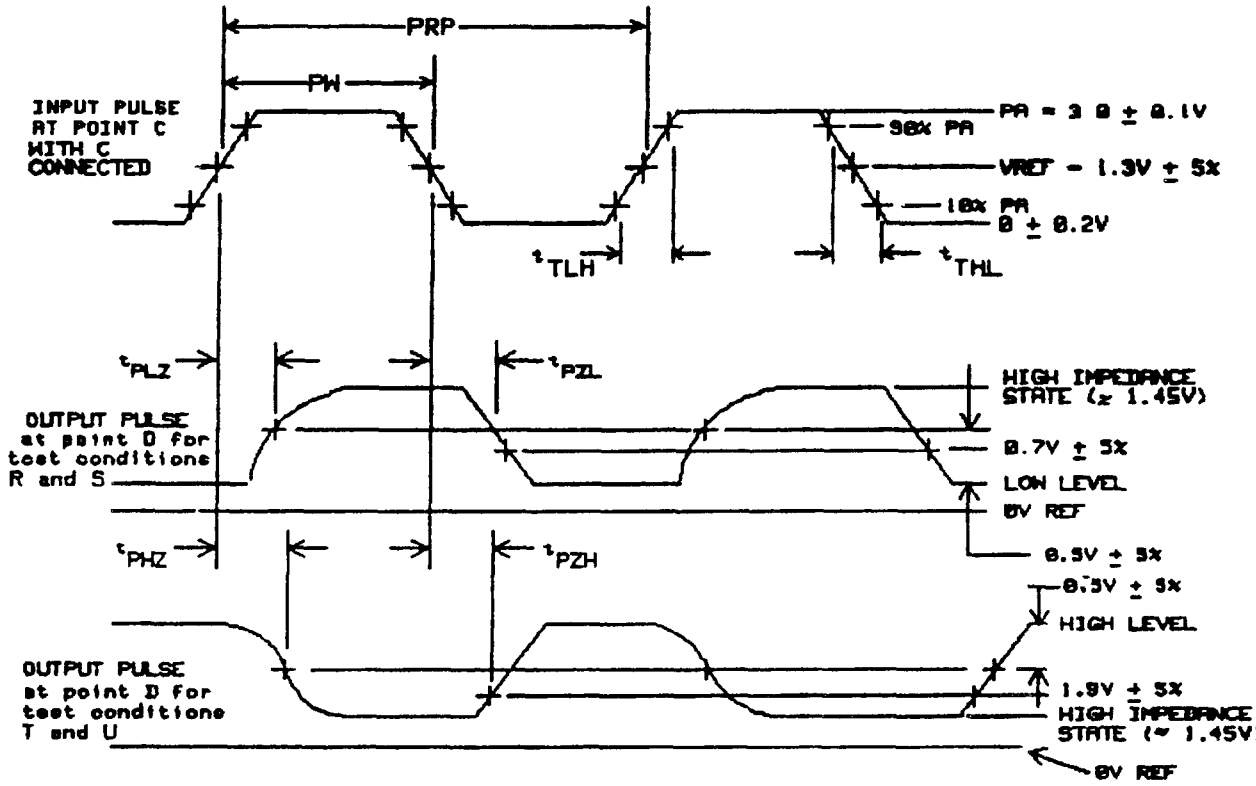
1. R1 = 51.1 Ohms ± 5 percent, 1/2 W.

2. Load the output under test with the load as specified on figure 13.

3. Abbreviations used are defined in 6.1.

FIGURE 18. Disable and enable delay time (tPLZ, tPZL, tPHZ, tPZH) test circuit for tables XVIII, XIX, XX and XXI.

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NOTES:

1. Input pulse setup:
   PW = 500 ± 50 nS
   PRP = 1000 ± 100 nS
   tTLH ≤ 15 nS, tTHL ≤ 6 nS.

2. Before making measurements, verify that the output high level is ≥ 2.4 V and the low level is ≤ 0.4 V.

3. All voltage levels are referenced to ground.

FIGURE 19. Disable and enable delay time (tPLZ, tPZL, tPHZ, tPZH) test circuit waveforms for tables XVIII, XIX, XX, XXI.

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TABLE XXII. Output crosstalk, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
V1	1Y1	2	Open	3,26,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	22	0	625	mV	20, 21, 13
V2	1Y2	3		2,26,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	23				
V3	1Y3	26		2,3,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	27				
V4	1Y4	24		2,3,26, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	25				
V5	2Y1	4		2,3,26, 24,6,31, 8,33,29, 35,13, 17,16, 39,38	5				
V6	2Y2	6	Open	2,3,26, 24,4,31, 8,33,29, 35,13, 17,16, 39,38	7	0	625	mV	20, 21, 13

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TABLE XXII. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
V7	2Y3	31	Open	2,3,26, 24,4,6, 8,33,29, 35,13, 17,16, 39,38	32	0	625	mV	20, 21, 13
V8	2Y4	8		2,3,26, 24,4,6, 31,33, 29,35, 13,17, 16,39,38	28				
V9	3Y1	33		2,3,26, 24,4,6, 31,8,29, 35,13, 17,16, 39,38	12				
V10	3Y2	29		2,3,26, 24,4,6, 31,8,33, 35,13, 17,16, 39,38	11				
V11	3Y3	35		2,3,26, 24,4,6, 31,8,33, 29,13, 17,16, 39,38	15				
V12	3Y4	13	Open	2,3,26, 24,4,6, 31,8,33, 29,35, 17,16, 39,38	34	0	625	mV	20, 21, 13

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Check the source to verify that this is the current version before use.

TABLE XXII. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
V13	4Y1	17	Open	2,3,26, 24,4,6, 31,8,33, 29,35, 13,16, 39,38	37	0	625	mV	20, 21, 13
V14	4Y2	16		2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 39,38	36				
V15	4Y3	39		2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 16,38	19				
V16	4Y4	38	Open	2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 16,39	18	0	625	mV	20, 21, 13

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TABLE XXIII. Output crosstalk, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
W1	1Y1	Open	2	3,26,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	22	0	1000	mV	20, 21, 13
W2	1Y2		3	2,26,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	23				
W3	1Y3		26	2,3,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	27				
W4	1Y4		24	2,3,26, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	25				
W5	2Y1		4	2,3,26, 24,6,31, 8,33,29, 35,13, 17,16, 39,38	5				
W6	2Y2	Open	6	2,3,26, 24,4,31, 8,33,29, 35,13, 17,16, 39,38	7	0	1000	mV	20, 21, 13

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TABLE XXIII. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
W7	2Y3	Open	31	2,3,26, 24,4,6, 8,33,29, 35,13, 17,16, 39,38	32	0	1000	mV	20, 21, 13
W8	2Y4		8	2,3,26, 24,4,6, 31,33, 29,35, 13,17, 16,39,38	28				
W9	3Y1		33	2,3,26, 24,4,6, 31,8,29, 35,13, 17,16, 39,38	12				
W10	3Y2		29	2,3,26, 24,4,6, 31,8,33, 35,13, 17,16, 39,38	11				
W11	3Y3		35	2,3,26, 24,4,6, 31,8,33, 29,13, 17,16, 39,38	15				
W12	3Y4	Open	13	2,3,26, 24,4,6, 31,8,33, 29,35, 17,16, 39,38	34	0	1000	mV	20, 21, 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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TABLE XXIII. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
W13	4Y1	Open	17	2,3,26, 24,4,6, 31,8,33, 29,35, 13,16, 39,38	37	0	1000	mV	20, 21, 13
W14	4Y2		16	2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 39,38	36				
W15	4Y3		39	2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 16,38	19				
W16	4Y4	Open	38	2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 16,39	18	0	1000	mV	20, 21, 13

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TABLE XXIV. Output crosstalk, duration, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
X1	1Y1	Open	2	3,26,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	22	0	33	nS	20, 22, 13
X2	1Y2		3	2,26,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	23				
X3	1Y3		26	2,3,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	27				
X4	1Y4		24	2,3,26, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	25				
X5	2Y1		4	2,3,26, 24,6,31, 8,33,29, 35,13, 17,16, 39,38	5				
X6	2Y2	Open	6	2,3,26, 24,4,31, 8,33,29, 35,13, 17,16, 39,38	7	0	33	nS	20, 22, 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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TABLE XXIV. Output crosstalk, duration, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
X7	2Y3	Open	31	2,3,26, 24,4,6, 8,33,29, 35,13, 17,16, 39,38	32	0	33	nS	20, 22, 13
X8	2Y4		8	2,3,26, 24,4,6, 31,33, 29,35, 13,17, 16,39,38	28				
X9	3Y1		33	2,3,26, 24,4,6, 31,8,29, 35,13, 17,16, 39,38	12				
X10	3Y2		29	2,3,26, 24,4,6, 31,8,33, 35,13, 17,16, 39,38	11				
X11	3Y3	Open	35	2,3,26, 24,4,6, 31,8,33, 29,13, 17,16, 39,38	15	0	33	nS	20, 22, 13

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TABLE XXIV. Output crosstalk, duration, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
X12	3Y4	Open	13	2,3,26, 24,4,6, 31,8,33, 29,35, 17,16, 39,38	34	0	33	nS	20, 22, 13
X13	4Y1		17	2,3,26, 24,4,6, 31,8,33, 29,35, 13,16, 39,38	37				
X14	4Y2		16	2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 39,38	36				
X15	4Y3		39	2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 16,38	19				
X16	4Y4	Open	38	2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 16,39	18	0	33	nS	20, 22, 13

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TABLE XXIV. Output crosstalk, duration, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
X17	1Y1	2	Open	3,26,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	22	0	33	nS	20, 22, 13
X18	1Y2	3		2,26,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	23				
X19	1Y3	26		2,3,24, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	27				
X20	1Y4	24		2,3,26, 4,6,31, 8,33,29, 35,13, 17,16, 39,38	25				
X21	2Y1	4		2,3,26, 24,6,31, 8,33,29, 35,13, 17,16, 39,38	5				
X22	2Y2	6	Open	2,3,26, 24,4,31, 8,33,29, 35,13, 17,16, 39,38	7	0	33	nS	20, 22, 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:58Z
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TABLE XXIV. Output crosstalk, duration, test - Continued.

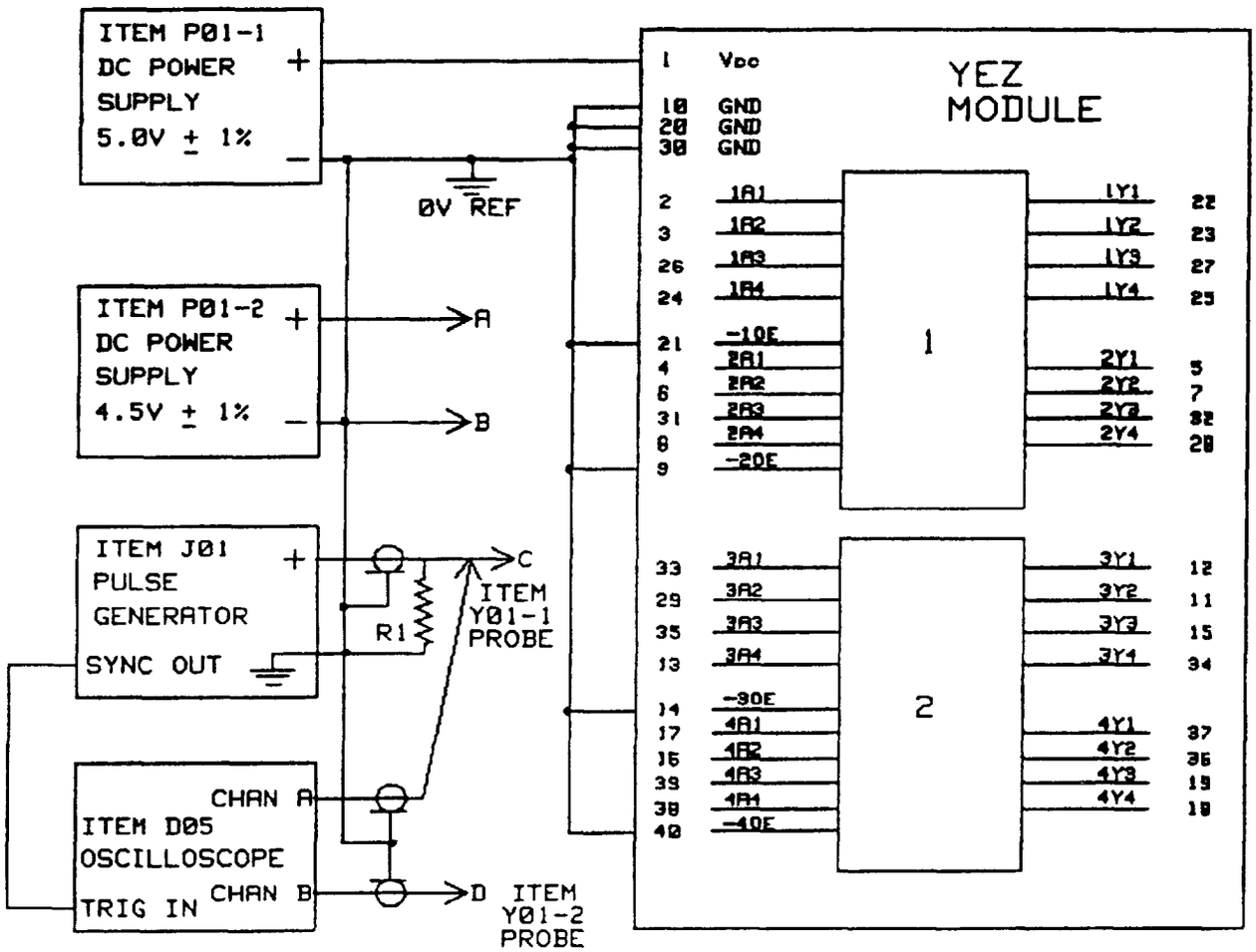
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
X23	2Y3	31	Open	2,3,26, 24,4,6, 8,33,29, 35,13, 17,16, 39,38	32	0	33	nS	20, 22, 13
X24	2Y4	8		2,3,26, 24,4,6, 31,33, 29,35, 13,17, 16,39,38	28				
X25	3Y1	33		2,3,26, 24,4,6, 31,8,29, 35,13, 17,16, 39,38	12				
X26	3Y2	29		2,3,26, 24,4,6, 31,8,33, 35,13, 17,16, 39,38	11				
X27	3Y3	35		2,3,26, 24,4,6, 31,8,33, 29,13, 17,16, 39,38	15				
X28	3Y4	13	Open	2,3,26, 24,4,6, 31,8,33, 29,35, 17,16, 39,38	34	0	33	nS	20, 22, 13

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TABLE XXIV. Output crosstalk, duration, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Test figs
X29	4Y1	17	Open	2,3,26, 24,4,6, 31,8,33, 29,35, 13,16, 39,38	37	0	33	nS	20, 22, 13
X30	4Y2	16		2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 39,38	36				
X31	4Y3	39		2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 16,38	19				
X32	4Y4	38	Open	2,3,26, 24,4,6, 31,8,33, 29,35, 13,17, 16,39	18	0	33	nS	20, 22, 13

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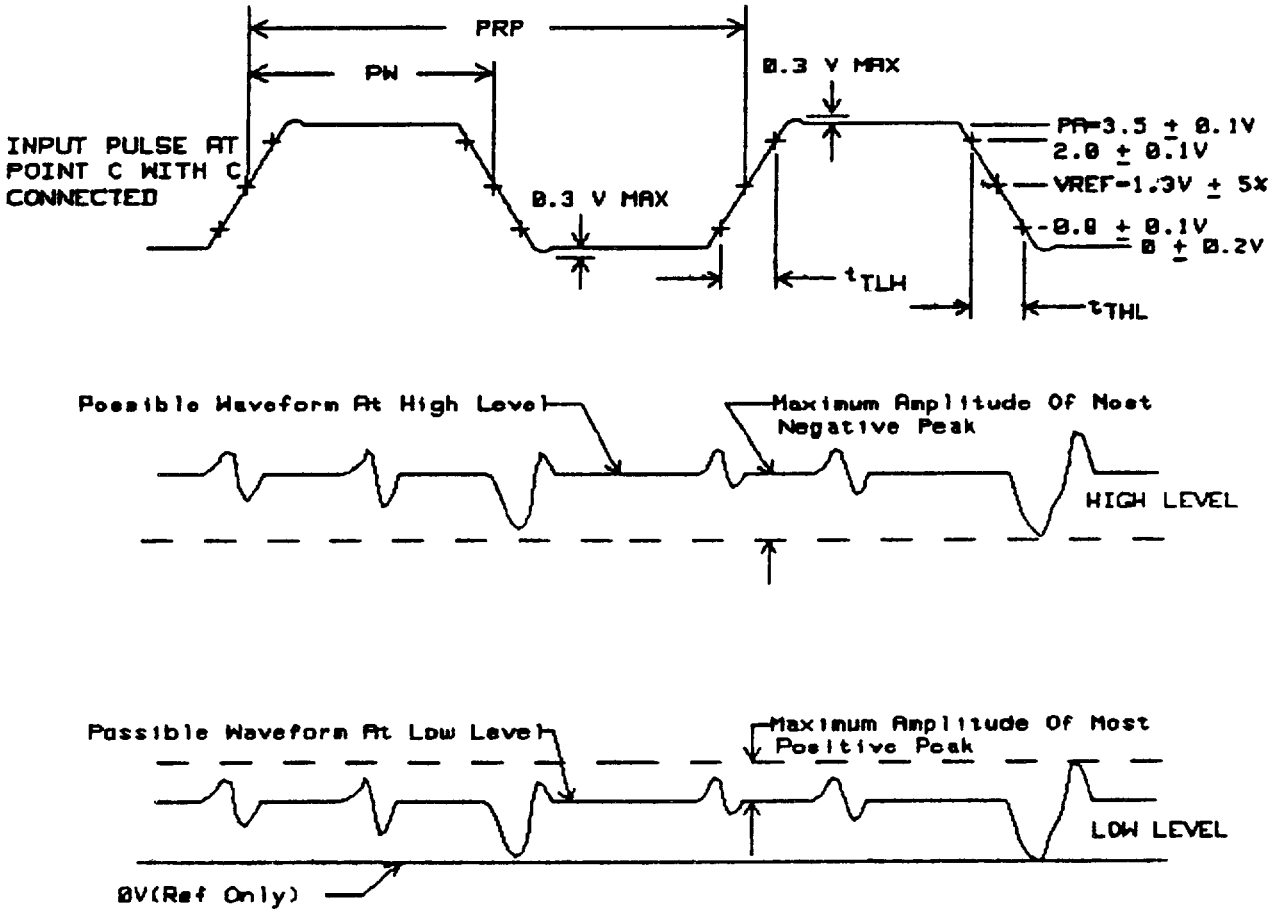


NOTES:

1. R1 = 51.1 Ohms ± 5 percent, 1/2W.
2. Load all the outputs with the load as specified on figure 13.
3. Abbreviations used are defined in 6.1.

FIGURE 20. Output crosstalk, high and low level and duration, test circuit for tables XXII, XXIII, and XXIV.

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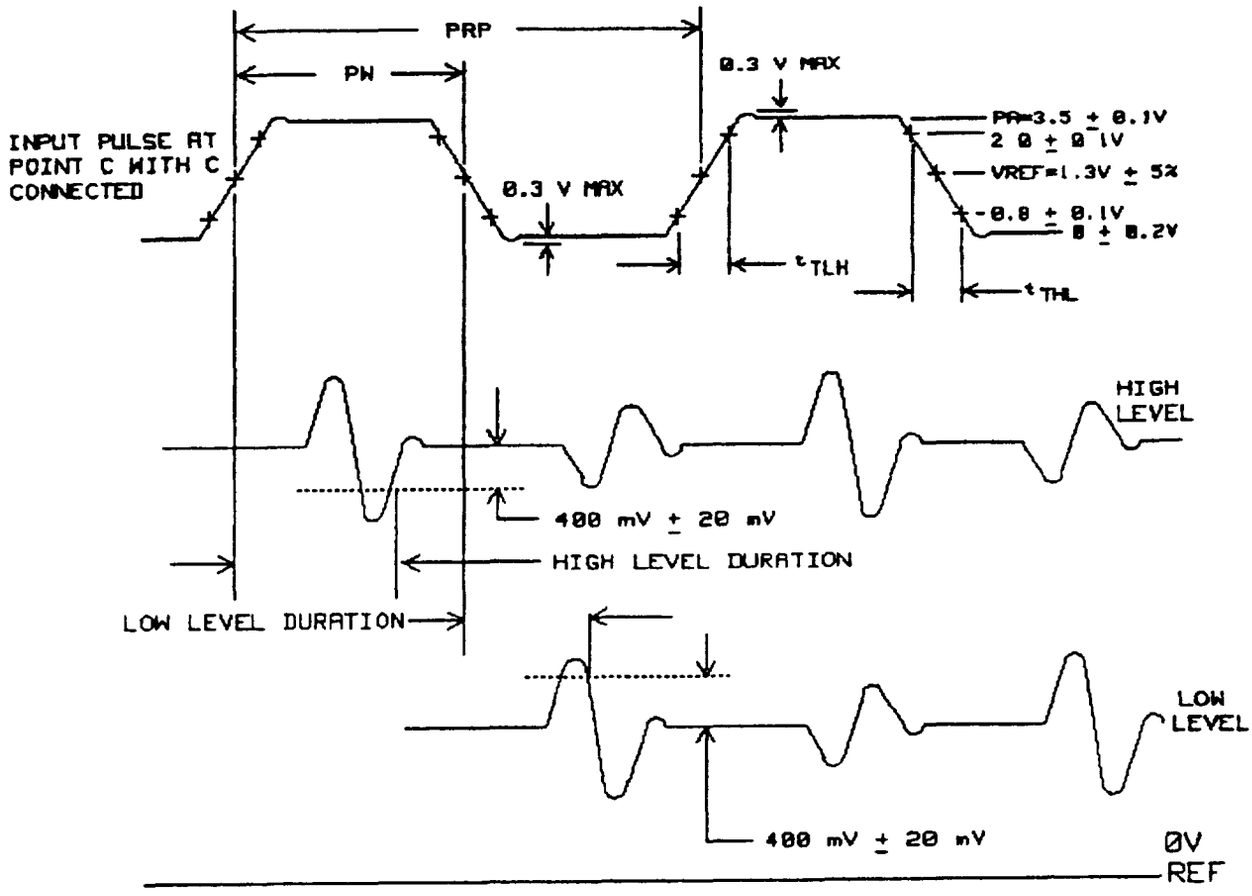


NOTE:

1. Input pulse setup:
   PW = 500 ± 50 nS
   PRP = 1000 ± 100 nS
   tTLH = tTHL = 3.5 ± 1.5 nS (verified at each output pin).

FIGURE 21. Output crosstalk, high and low level, test circuit waveforms for tables XXII and XXIII.

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NOTES:

1. Input pulse setup:
   PW = 500 ± 50 nS.
   PRP = 1000 ± 100 nS.
   tTLH = tTHL = 3.5 ± 1.5 nS (verified at each output pin).

2. The output crosstalk duration is measured from the input pulse to the point where the output crosstalk remains below the specified voltage. If the output crosstalk amplitude is less than the specified reference voltage, the output crosstalk duration shall be recorded as 0 nS.

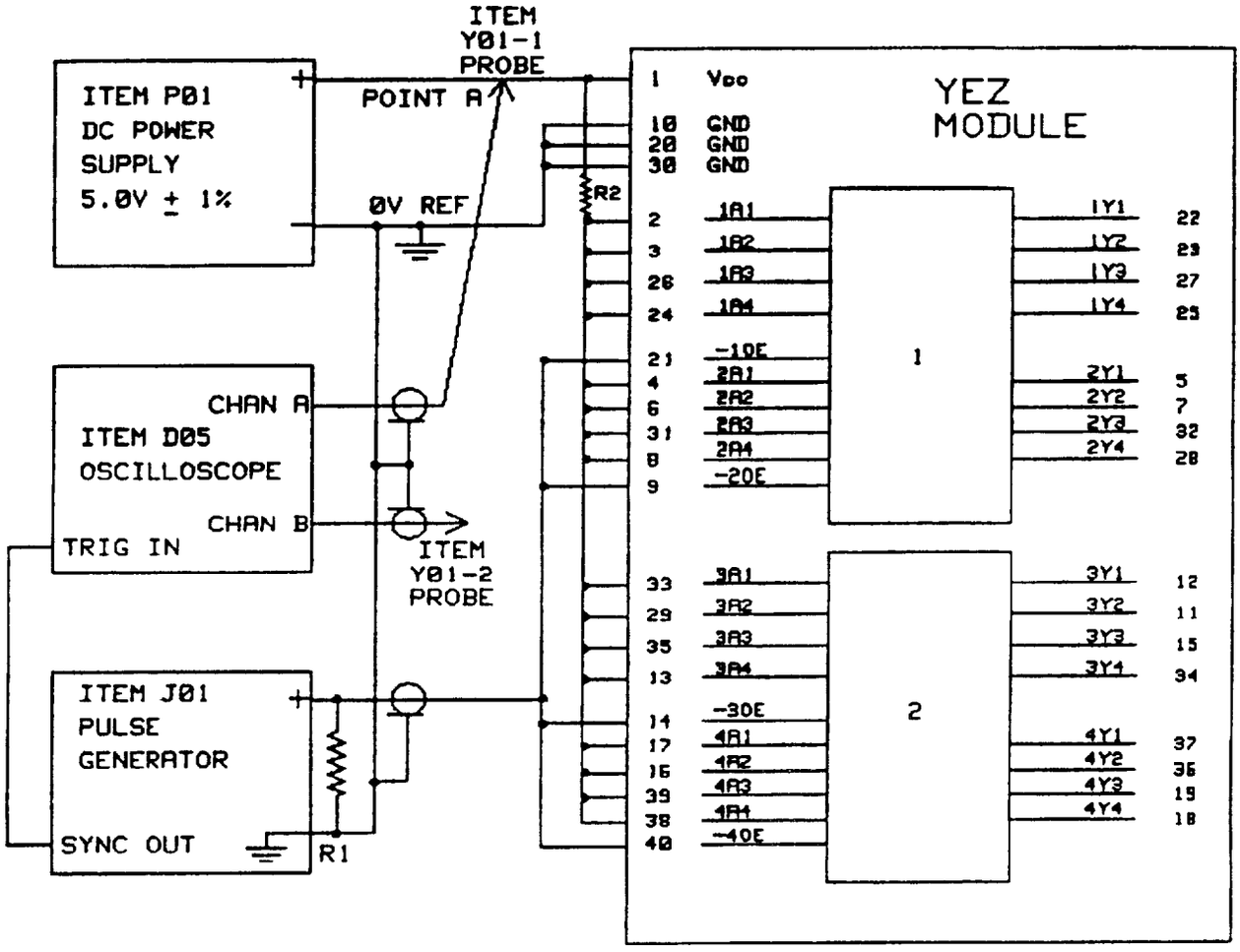
FIGURE 22. Output crosstalk duration waveforms for table XXIV.

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TABLE XXV. Power supply transient amplitude test.

Test cond	Parameter	25 ± 5°C Initial limits		Unit	Test figs
		Min	Max		
Y	PSTA	0	400	mV	23, 24

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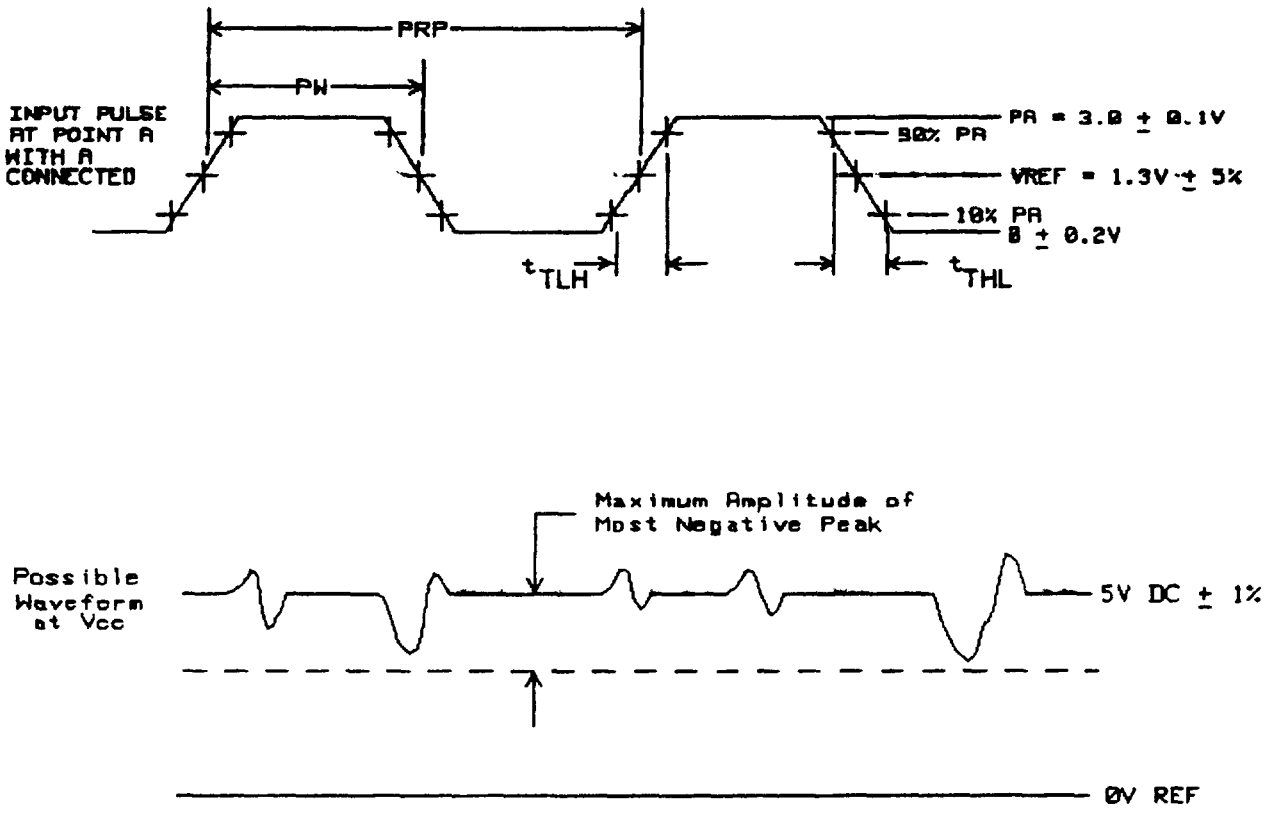


NOTES:

1. R1 = 51.1 Ohms ± 5 percent, 1/2 W.
2. R2 = 1.0 KOhm ± 5 percent, 1/4 W.
3. Abbreviations used are defined in 6.1.

FIGURE 23. Power supply transient amplitude test circuit for table XXV.

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NOTE:

1. Input pulse setup:
   PW = 500 ± 50nS
   PRP = 1000 ± 100 nS
   tTLH ≤ 15 nS, tTHL ≤ 6 nS

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XXV.

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TABLE XXVI. Power supply filter capacitance test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Unit	Test fig
		A to pin	B to pin	Min	Max		
Z	PSFC	1	Open	3.83	6.38	uF	25

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TABLE XXVII. Input capacitance test.

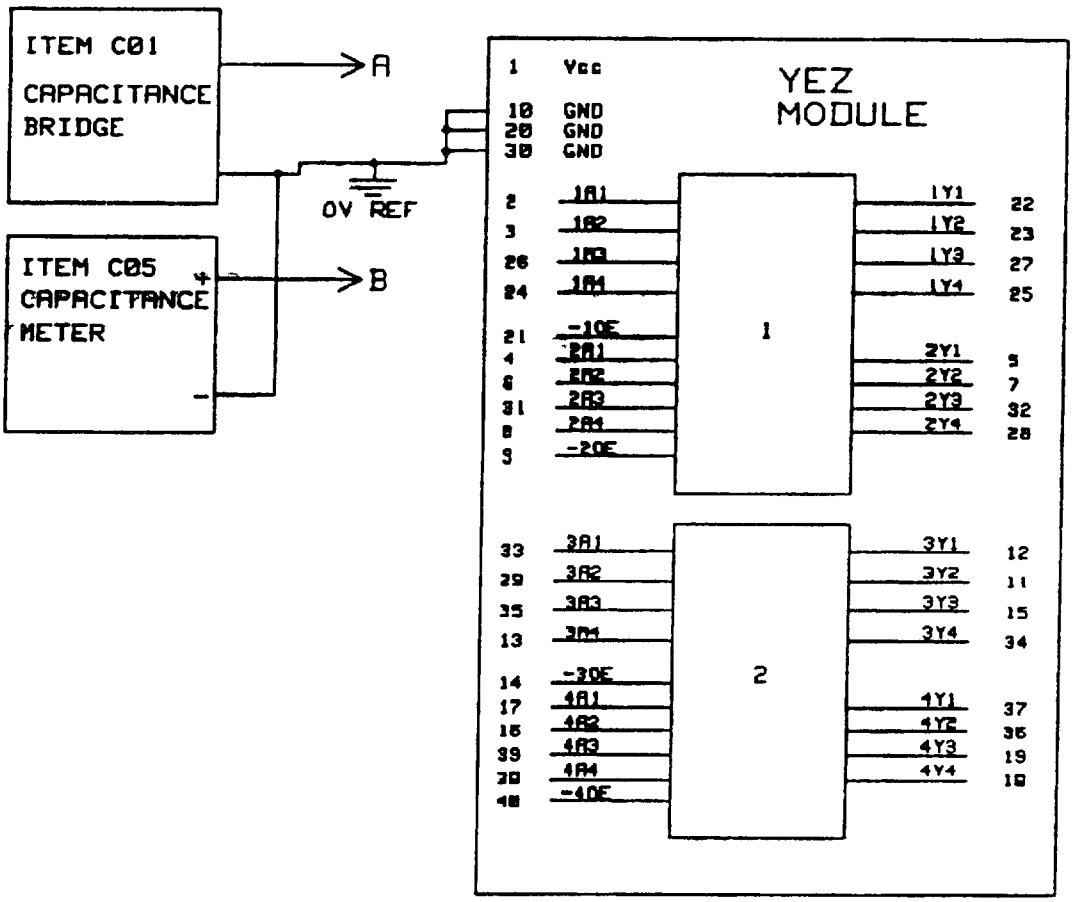
Test cond	Signal name	Connect		25 ± 5°C Initial limits		Unit	Test fig
		A to pin	B to pin	Min	Max		
AA1	1A1	Open	2	2	30	pF	25
AA2	1A2		3				
AA3	1A3		26				
AA4	1A4		24				
AA5	2A1		4				
AA6	2A2		6				
AA7	2A3		31				
AA8	2A4		8				
AA9	3A1		33				
AA10	3A2		29				
AA11	3A3		35				
AA12	3A4		13				
AA13	4A1		17				
AA14	4A2		16				
AA15	4A3		39				
AA16	4A4		38	2	30	pF	25
AA17	-1OE		21	2	35	pF	25
AA18	-2OE		9				
AA19	-3OE		14				
AA20	-4OE	Open	40	2	35	pF	25

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TABLE XXVIII. Output capacitance test.

Test cond	Signal name	Connect		25 ± 5°C Initial limits		Unit	Test fig
		A to pin	B to pin	Min	Max		
BB1	1Y1	Open	22	2	35	pF	25
BB2	1Y2		23				
BB3	1Y3		27				
BB4	1Y4		25				
BB5	2Y1		5				
BB6	2Y2		7				
BB7	2Y3		32				
BB8	2Y4		28				
BB9	3Y1		12				
BB10	3Y2		11				
BB11	3Y3		15				
BB12	3Y4		34				
BB13	4Y1		37				
BB14	4Y2		36				
BB15	4Y3		19				
BB16	4Y4	Open	18	2	35	pF	25

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NOTES:

1. Null out the effect of the test leads prior to connecting to the module.

2. Do not attach test connector or wires to any of the module pins other than the pin under test. Stray capacitance added to other pins may affect the readings.

3. Capacitanoe bridge (C01) setup:
   Frequency = 1 kHz
   Vpk = 0.3V maximum.

4. Abbreviations used are defined in 6.1.

FIGURE 25. Power supply filter, input and output capacitance test circuit for tables XXVI, XXVII, and XXVIII.

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